

Randomized Resource Allocation in Decentralized Wireless Networks

Kamyar Moshksar, *Student Member, IEEE*, Alireza Bayesteh, *Member, IEEE*, and Amir K. Khandani, *Member, IEEE*

Abstract—In this paper, we consider a decentralized wireless communication network with a fixed number u of frequency subbands to be shared among N transmitter-receiver pairs. It is assumed that the number of active users is a realization of a random variable with a given probability mass function. Moreover, users are unaware of each other's codebooks and hence, no multiuser detection is possible. We propose a randomized frequency hopping (FH) scheme in which each transmitter randomly hops over a subset of u subbands from transmission slot to transmission slot. Assuming all users transmit Gaussian signals, the distribution of the noise plus interference is mixed Gaussian, which makes calculation of the mutual information between the transmitted and received signals of each user intractable. We derive lower and upper bounds on the mutual information of each user and demonstrate that, for large signal-to-noise ratio (SNR) values, the two bounds coincide. This observation enables us to compute the sum multiplexing gain of the system and obtain the optimum hopping strategy for maximizing this quantity. We compare the performance of the FH system to that of the frequency division (FD) system in terms of the following performance measures: average sum multiplexing gain ($\eta^{(1)}$) and average minimum multiplexing gain per user ($\eta^{(2)}$). We show that (depending on the probability mass function of the number of active users) the FH system can offer a significant improvement in terms of $\eta^{(1)}$ and $\eta^{(2)}$ (implying a more efficient usage of the spectrum). In the sequel, we consider a scenario where the transmitters are unaware of the number of active users in the network as well as the channel gains. Developing a new upper bound on the differential entropy of a mixed Gaussian random vector and using entropy power inequality, we obtain lower bounds on the maximum transmission rate per user to ensure a specified outage probability at a given SNR level. We demonstrate that the so-called outage capacity can be considerably higher in the FH scheme than in the FD scenario for reasonable distributions on the number of active users. This guarantees a higher spectral efficiency in FH compared to FD.

Index Terms—Decentralized networks, frequency hopping (FH), mixed Gaussian interference, multiplexing gain, outage capacity, randomized signaling, spectrum sharing.

Manuscript received December 02, 2009; revised August 08, 2010; accepted August 18, 2010. Date of current version March 16, 2011. This work was supported by research grants provided by Natural Sciences and Engineering Council of Canada (NSERC). The material in this paper was presented at the International Symposium in Information Theory (ISIT08) Toronto, ON, Canada, July 2008.

K. Moshksar and A. K. Khandani are with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON N2L 3G1 Canada (e-mail: khandani@shannon2.uwaterloo.ca; kmoshksar@cst.uwaterloo.ca; khandani@cst.uwaterloo.ca).

A. Bayesteh is with Research in Motion, Waterloo, ON N2L 5Z5 Canada (e-mail: abayesteh@rim.com).

Communicated by M. Franceschetti, Associate Editor for Communication Networks.

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TIT.2011.2112030

I. INTRODUCTION

A. Motivation

INCREASING demand for wireless applications on one hand, and the limited available resources on the other hand, provoke more efficient usage of such resources. Due to its significance, many researchers have addressed the problem of resource allocation in wireless networks. One major challenge in wireless networks is the destructive effect of multiuser interference, which degrades the performance when multiple users share the spectrum. As such, an efficient and low complexity resource allocation scheme that maximizes the quality of service while mitigating the impact of the multiuser interference is desirable. The existing resource allocation schemes are either *centralized*, i.e., a central controller manages the resources, or *decentralized*, where resource allocation is performed locally at each node. Due to the complexity of adapting the centralized schemes to the network structure (e.g., number of active users), these schemes are usually designed for a fixed network structure. This makes inefficient usage of resources because, in most cases, the number of active users may be considerably less than the value assumed in the design process. On the other hand, most of the decentralized resource allocation schemes suffer from the complexity, either in the algorithm (e.g., game-theoretic approaches involving iterative methods) or in the hardware (e.g., cognitive radio). Therefore, it is of interest to devise an efficient and low-complexity decentralized resource allocation scheme, which is the main goal of this paper.

B. Related Works

1) Centralized Schemes: In recent years, many centralized power and spectrum allocation schemes have been studied in cellular and multihop wireless networks [1]–[8]. Clearly, centralized schemes perform better than the decentralized (distributed) approaches, while requiring extensive knowledge of the network configuration. In particular, when the number of nodes is large, deploying such centralized schemes may not be practically feasible.

Traditional wireless systems aimed to avoid the interference among users by using orthogonal transmission schemes. The most common example is the frequency division (FD) system, in which different users transmit over disjoint frequency subbands. The assignment of frequency subbands is usually performed by a central controller. Despite its simplicity, FD is shown to achieve the highest throughput in certain scenarios. In particular, [9] proves that in a wireless network where interference is treated as noise (no multiuser detection is performed), if the crossover gains are sufficiently larger than the forward gains, FD is Pareto-rate-optimal. Due to practical considerations, such FD

systems usually rely on a fixed number of frequency subbands. Hence, if the number of users changes, the system is not guaranteed to offer the best possible spectral efficiency because, most of the time, the majority of the potential users may be inactive.

2) *Decentralized Schemes*: In decentralized schemes, decisions concerning network resources are made by individual nodes based on their local information. Most of decentralized schemes reported in the literature rely on either *game-theoretic* approaches or *cognitive radios*. Cognitive radios [10] have the ability to sense the unoccupied portion of the available spectrum and use this information in resource allocation. Fundamental limits of wireless networks with cognitive radios are studied in [11]–[14]. Although cognitive radios avoid the use of a central controller, they require sophisticated detection techniques for sensing the spectrum holes and dynamic frequency assignment, which add to the overall system complexity [15]. Noting the above points, it is desirable to have a decentralized frequency sharing strategy without the need for cognitive radios, which allows the users to coexist while utilizing the spectrum efficiently and fairly.

Being a standard technique in spread spectrum communications and due to its interference avoidance nature, hopping is the simplest spectrum sharing method to use in decentralized networks. As different users typically have no prior information about the codebooks of the other users, the most efficient method is avoiding interference by choosing unused channels. As aforementioned, searching the spectrum to find spectrum holes is not an easy task due to the dynamic spectrum usage. As such, frequency hopping (FH) is a realization of a transmission scheme without sensing, while avoiding the collisions as much as possible. FH is one of the standard signaling schemes [16] adopted in *ad hoc* networks. In short range scenarios, bluetooth systems [17]–[19] are the most popular examples of a wireless personal area network (WPAN). Using FH over the unlicensed ISM band, a bluetooth system provides robust communication to unpredictable sources of interference. A modification of FH, called dynamic frequency hopping (DFH), selects the hopping pattern based on interference measurements in order to avoid dominant interferers. The performance of a DFH scheme when applied to a cellular system is assessed in [20]–[22].

Although there has been a tremendous amount of work on the performance evaluation of hopping-based decentralized networks, there are only a few information-theoretic results reflecting the fundamental limits of such networks. This paper offers a thorough analysis of a decentralized network where all users follow a randomized hopping strategy to share the resources, while the number of active users is a random variable with a given distribution. In the first part of the paper, the channel gains and the number of active users are assumed to be static and known to the corresponding transmitter-receiver nodes. This assumption makes the concept of achievable rate in the Shannon sense meaningful. However, in case the channel gains and the number of active users are not known to the transmitters, the concept of achievable rate may no longer be valid. A common setup for such an assumption is a network where channel gains are quasi-static fading and unknown to the transmitters, which is the framework for the last part of the paper.

Rayleigh fading is an unavoidable phenomenon in wireless networks that can affect the performance of the system signif-

icantly. Traditionally, Rayleigh fading has been considered to be harmful due to reducing the transmission reliability in wireless networks. However, recently, researchers have been able to reduce this harmful effect by exploiting the so-called *multiuser diversity* [23], [24]. This can be considered a *scheduling gain* by allowing the users with favorable channels to be active. It is shown that multiuser diversity gain can be as large as $\log \log N$ in broadcast and multiple-access channels [25]–[27] and $\log N$ in single-hop ad hoc networks [28], [29], as N grows to infinity. However, achieving this scheduling gain requires the fading channels to vary over time such that all possible realizations of the fading process are covered. In the case that channel gains are selected randomly at the start of the transmission and remain constant during the whole transmission period (quasi-static fading), the channels do not have ergodic behavior. In this case, a suitable performance measure is the so-called ε - *outage capacity* [30], denoted by $R(\varepsilon)$, which is defined as the maximum transmission rate per user, ensuring an outage probability below ε , i.e.

$$R(\varepsilon) = \sup\{R : \Pr\{\text{Outage}\} < \varepsilon\}.$$

The reality of wireless channels is more complicated to be simply represented by the Rayleigh fading model. A class of channel models considered in the literature is the one in which the signal power decays according to a distance-based attenuation law [31]–[38]. Moreover, the presence of obstacles adds some randomness (known as shadowing) to the received signal. It is well known that the effects of such random phenomena can significantly affect the throughput of a spectrum sharing network in both multi-hop [39]–[42] and single-hop scenarios [43, Ch. 8], [44]–[48]. These features indeed increase the frequency reuse factor as they will attenuate the interference caused by a given transmitter on its neighboring receivers. In spite of the significance of the effects of distance-based attenuation and shadowing on the throughput of a spectrum sharing system, unfortunately, there is not a single commonly accepted model to represent these factors. It should be emphasized that the inclusion of signal attenuation due to distance and/or shadowing will indeed simplify the spectrum sharing as the multiuser interference will be attenuated and consequently its harmful effect will be reduced. On the other hand, these factors do not impact the performance of the orthogonal schemes in which the multiuser interference is altogether avoided. In spite of this fact, as the actual model used to represent distance-based attenuation and shadowing can have a profound impact on the system throughput (to the advantage of the randomized spectrum sharing schemes advocated in the current article), to avoid any confusion, we have relied on a simple Rayleigh fading model which in some sense captures the minimum advantage offered by the proposed scheme versus those based on orthogonal separation of users.

Reference [49] studies a wireless network composed of a set of transmitter/receiver pairs in which a given link can be off or transmit with a constant power. Both the case of Rayleigh fading as well as a Rayleigh fading mixed with a proper distance-based attenuation are considered and a comparison between the scaling (with respect to the number of links) of the throughput in these two cases is provided.

In [50] and [51], the authors study a decentralized wireless ad hoc network where different transmitters are connected to

different receivers through channels with a similar path loss exponent. Assuming the transmitters are scattered over the two dimensional plane according to a Poisson point process, a fixed bandwidth is partitioned into a certain number of subbands, such that the so-called transmission intensity in the network is maximized, while the probability of outage per user is below a certain threshold [50]. The transmission strategy is based on choosing one subband randomly per transmission, which is a special case of FH. In [51], a noniterative and distributed power control scheme is introduced, for which the constant power and the channel inversion schemes are extreme cases. It is observed that none of these cases are ideal in general. In fact, it is shown that regulating the transmission power proportionally to the inverse square root of the forward channel strength minimizes the outage probability.

FH is also proposed in [14] in the context of cognitive radios, where each cognitive transmitter selects a frequency subband but quits transmitting if the subband is already occupied by a primary user.

The popularity of FH motivates us to consider this scheme operating over a fixed bandwidth where due to practical constraints, the spectrum is divided to a number u of frequency subbands. We note that the results of the paper are valid in a general setup where hopping is performed over an arbitrary orthogonal basis. To make the presentation as simple as possible, we take the orthogonal basis in the frequency domain.

We close this section by mentioning that the long-known ALOHA system [52] is a potential candidate for data communication in a decentralized network. In an ALOHA-like system, each user randomly selects a number of frequency subbands upon activation and transmits the entire symbols in its codeword through the selected subbands. Therefore, if two users overlap (partially or completely) on their selected subbands, they lose (some or the entire) information in the transmitted packets. However, in the randomized FH scheme, each user transmits the different symbols in its codeword through randomly selected subbands which changes from transmission slot to transmission slot. Overall, in an ALOHA-like system there is a nonzero probability that any user loses its entire codeword regardless of the transmission rate, while in the randomized FH scenario, one can always transmit below a certain rate and recover its data completely for each transmission block. Although, the average throughput of an ALOHA-like system can be shown to be the same as that of the randomized FH scheme in the high SNR regime, there are basic assumptions made in an ALOHA system that are not necessary in the FH scheme:

- 1 The need to retransmit the lost packets in ALOHA-like systems demands the receiver to notify its associated transmitter that a packet is lost. This leads to considerable delay and stability¹ issues regarding the backlogged packets [53]–[55]. Also, there must be a feedback link from any receiver to its affiliated transmitter in order to report the lost packets. This decreases the overall spectral efficiency.
- 2 In the ALOHA system, all users are assumed to be block and symbol synchronized. However, in the proposed FH

¹An ALOHA system is called stable if the Markovian process of backlogged packets is ergodic.

scheme, the assumption on block synchronization is either unnecessary (blocks are transmitted without interruption) or the analysis can be easily generalized to include cases that there are periods of silence between blocks.

Recently, researchers have aimed to improve the average throughput of an ALOHA network in the context of random multiple access. For example, let there be at most two users in the network where each user is active with a probability p . Inspired by the fact that the capacity region of a two-user multiple access channel is a polytope with two corner points (τ_1, τ_2) and (τ'_1, τ'_2) where τ_1, τ_2, τ'_1 , and τ'_2 are nonzero rates, the authors in [56] propose that user #1 (user #2) selects τ_1 and τ'_1 (τ_2 and τ'_2) with probabilities q and $1 - q$, respectively. As far as the pair of selected rates lies in the capacity region of the corresponding multiple access channel, the messages sent by both users can be received with arbitrarily small probability of error. Finally, q is selected to maximize the average throughput. An elegant approach is proposed in [57] where users transmit over several data streams (using message splitting) and the receiver decodes only certain data streams depending on which users are active. This results in characterizing the capacity region of the Gaussian random multiple access channel within a constant gap in a scenario where the activation status of both users is only available at the receiver. We emphasize that these techniques do not apply in the setup of a decentralized network as the capacity region of a decentralized interference channel is unknown and multiuser detection is not possible.

C. Contribution

In this paper, we consider a decentralized wireless communication network with a fixed number u of frequency subbands to be shared among N transmitter-receiver pairs. It is assumed that the number of active users is a realization of a random variable with a given probability mass function. Moreover, users are unaware of each other's codebooks, and hence, no multiuser detection is possible. We propose a randomized FH scheme in which the i th transmitter randomly hops over v_i (the hopping parameter) out of u subbands from transmission slot to transmission slot. Assuming i.i.d. Gaussian signals are transmitted over the chosen subbands, the distribution of the noise plus interference becomes mixed Gaussian, which makes the calculation of the achievable rate complicated. The main contributions of the paper are presented in two different cases.

Case I: Transmitters are Aware of the Number of Active Users and the Channel Gains:

- We derive lower and upper bounds on the mutual information between the transmitted and received signals of each user and demonstrate that, for large SNR values, the two bounds coincide. Thereafter, we are able to show that the achievable rate of the i th user scales like $\frac{v_i}{2} \prod_{j=1, j \neq i}^N (1 - \frac{v_j}{u}) \log \text{SNR}$.
- We show that each transmitter only needs the knowledge of the number of active users in the network, the forward channel gain and the maximum interference level at its associated receiver to regulate its transmission rate. Knowing these quantities, we demonstrate how the i th user can achieve a multiplexing gain of $\frac{v_i}{2} \prod_{j=1, j \neq i}^N (1 - \frac{v_j}{u})$.

- We obtain the optimum design parameters $\{v_i\}_{i=1}^N$ in order to maximize various performance measures.
- We compare the performance of the FH with that of the FD in terms of the following performance measures: average sum multiplexing gain ($\eta^{(1)}$) and average minimum multiplexing gain per user ($\eta^{(2)}$). We show that (depending on the probability mass function of the number of active users) the FH system can offer a significant improvement in terms of $\eta^{(1)}$ and $\eta^{(2)}$ (implying a more efficient usage of the spectrum).

Case II: Transmitters are Unaware of the Number of Active Users and the Channel Gains:

- Developing a new upper bound on the differential entropy of a class of mixed Gaussian random vectors and using entropy power inequality, we offer three lower bounds on the ε -outage capacity of each user denoted by $R_{\text{FH}}^{(1)}(\varepsilon)$, $R_{\text{FH}}^{(2)}(\varepsilon)$, and $R_{\text{FH}}^{(3)}(\varepsilon)$. To evaluate the system performance analytically, we use $R_{\text{FH}}^{(3)}(\varepsilon)$, which can be computed easily. However, computation of $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$ involves integrations that cannot be carried out in a closed form. In the simulation results, we use the lower bounds $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$, which are tighter than $R_{\text{FH}}^{(3)}(\varepsilon)$.
- We perform asymptotic analysis for the outage capacity in terms of ε and SNR. In the asymptotically small ε regime, we observe that the maximum of outage capacity is obtained when the hopping parameter for each user is either 1 or u . In the asymptotically small SNR regime, we demonstrate that the system achieves the optimal performance regardless of the value of the hopping parameter. For asymptotically large values of SNR, it is shown that the optimal hopping parameter is $\left\lceil \frac{u}{n_{\max}} \right\rceil$, where $n_{\max}$ is the maximum possible number of concurrently active users in the network.
- We compare the outage capacity of the underlying FH scenario with that of the FD scheme for various scenarios in terms of distributions on the number of active users, SNR and ε . It is shown that FH outperforms FD in terms of outage capacity in many cases. We observe that in the low SNR regime, FH and FD offer the same performance. In the low ε regime, FD is always better than FH, however, for many practical scenarios there exists a threshold ε_{th} , such that FH outperforms FD as far as $\varepsilon \geq \varepsilon_{\text{th}}$. Also, we have shown that supremacy of FH over FD in the high SNR regime occurs quite often.

The paper outline is as follows. The system model is given in Section II. Section III offers an analysis of the achievable rates. Upper bounds and lower bounds on the achievable rates of users are presented in this section. In Section IV, based on the results in Section III, we discuss how users in the FH system can fairly share the spectrum. Comparison with the FD scheme in terms of the performance measures $\eta^{(1)}$ and $\eta^{(2)}$ is discussed in Section V. Section VI offers a comparison between two versions of the proposed randomized FH scheme, i.e., the robust FH and adaptive FH. Section VII describes analysis of the outage capacity. Derivation of various lower bounds on the outage capacity of users and comparison between the FH and FD scenarios through simulation results are part of the materials in this section. Finally, Section VIII concludes the paper.

D. Notation

Throughout the paper, we use the notation $E\{\cdot\}$ for the expectation operator. For a function $f(X, Y)$ of two independent random variables X and Y , $E_X\{f(X, Y)\}$ denotes the expectation of $f(X, Y)$ with respect to X while Y is treated as a parameter. $\Pr\{\mathfrak{E}\}$ denotes the probability of an event $\mathfrak{E}$, $\mathbb{1}(\mathfrak{E})$ the indicator function of an event $\mathfrak{E}$ and $p_X(\cdot)$ the probability density function (pdf) of a random variable X . Also, $I(X; Y)$ denotes the mutual information between random variables X and Y , $h(X)$ the differential entropy of a continuous random variable X and $H(X)$ the entropy of a discrete random variable X . The binary entropy function is denoted by $\mathcal{H}(\cdot)$, i.e., $\mathcal{H}(a) = -a \log a - (1 - a) \log(1 - a)$ for $a \in [0, 1]$. The pdf of a real jointly Gaussian vector of size $t \times 1$ and covariance matrix C is denoted by $g_t(\cdot, C)$. This notation is used even if C is singular. For example, if $t = 2$ and $C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, then $g_2(x_1, x_2, C) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x_1^2}{2}\right) \delta(x_2)$ where $\delta(\cdot)$ is the Dirac's delta function. The set of positive real numbers is denoted by $\mathbb{R}^+$. The conjugate transpose of a vector $\vec{x}$ is denoted by $\vec{x}^H$. The notation $f(x) \sim g(x)$ implies $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$. The notation $f(x) \approx g(x)$ means that $f(x)$ and $g(x)$ are both positive or both negative and $\lim_{x \rightarrow a} \frac{|f(x) - g(x)|}{\min\{|f(x)|, |g(x)|\}} = 0$ where a is clear from the context. Finally, $\triangleq$ means "is defined by."

II. SYSTEM MODEL AND ASSUMPTIONS

We consider a wireless network with N users² operating over a spectrum consisting of u orthogonal subbands. The number of active users is assumed to be a realization of a random variable with a given distribution, however, it is fixed during the whole transmission once it is set first. The transmission blocks of each user comprise of an arbitrarily large number of transmission slots. We remark that the results of this paper are valid regardless of having block synchronization among the users, however, we assume synchronization at the symbol level. The i th user exploits $v_i (\leq u)$ out of the u subbands in each transmission slot and hops randomly to another set of v_i frequency subbands in the next transmission slot. This user transmits independent real Gaussian signals of variance $\frac{P}{v_i}$ over the chosen subbands, in which P denotes the total average power for each transmitter. Each receiver is assumed to know the hopping pattern of its affiliated transmitter. It is assumed that the users are not aware of each other's codebooks and, hence, no multiuser detection or interference cancelation is possible at the receiver sides. The static and nonfrequency selective channel gain of the link connecting the i th transmitter to the j th receiver is shown by $h_{i,j}$. As it will be shown in (42), the only information each transmitter needs in order to regulate its transmission rate (focusing on the achieved multiplexing gain) is its forward channel gain, the maximum interference level at its associated receiver and the number of active users in the network. This information can be obtained at the receiver side and provided to the corresponding transmitter via a feedback link.³

²Each user consists of a separate transmitter-receiver pair.

³This is addressed in Section V.

As all users hop over different portions of the spectrum from transmission slot to transmission slot, no user is assumed to be capable of tracking the instantaneous interference level. This assumption makes the interference plus noise pdf at the receiver side of each user be mixed Gaussian. In fact, depending on different choices the other users make to select the frequency subbands and values of the crossover gains, the interference on each frequency subband at any given receiver can have up to 2^{N-1} power levels. The vector consisting of the received signals on the frequency subbands at the i th receiver in a typical transmission slot is

$$\vec{Y}_i = h_{i,i}\vec{X}_i + \vec{Z}_i \quad (1)$$

where $\vec{X}_i$ is the $u \times 1$ transmitted vector and $\vec{Z}_i$ is the noise plus interference vector at the receiver side of the i th user. One may write $p_{\vec{X}_i}(\cdot)$ as

$$p_{\vec{X}_i}(\vec{x}) = \sum_{C \in \mathcal{C}} \frac{1}{\binom{u}{v_i}} g_u(\vec{x}, C) \quad (2)$$

which corresponds to the mixed Gaussian distribution. In the above equation, the set $\mathcal{C}$ includes all $u \times u$ diagonal matrices in which v_i out of the u diagonal elements are $\frac{P}{v_i}$ and the rest are zero. Denoting the noise plus interference on the j th subband at the receiver side of the i th user by $Z_{i,j}$ (the j th component of $\vec{Z}_i$), it is clear that $p_{Z_{i,j}}(\cdot)$ is not dependent on j . This is due to the fact that the crossover gains are not frequency selective and there is no particular interest in a specific frequency subband by any user. We assume there are $L_i + 1$ ($L_i \leq 2^{N-1} - 1$) possible nonzero power levels for $Z_{i,j}$, say $\{\sigma_{i,l}^2\}_{l=0}^{L_i}$. Denoting the occurrence probability of $\sigma_{i,l}^2$ by $a_{i,l}$, $p_{Z_{i,j}}(\cdot)$ identifies a mixed Gaussian pdf as

$$p_{Z_{i,j}}(z) = \sum_{l=0}^{L_i} \frac{a_{i,l}}{\sqrt{2\pi}\sigma_{i,l}} \exp\left(-\frac{z^2}{2\sigma_{i,l}^2}\right) \quad (3)$$

where $\sigma^2 = \sigma_{i,0}^2 < \sigma_{i,1}^2 < \sigma_{i,2}^2 < \dots < \sigma_{i,L_i}^2$ (σ^2 is the ambient noise power). We notice that for each $l \geq 0$, there exists a $c_{i,l} \geq 0$ such that $\sigma_{i,l}^2 = \sigma^2 + c_{i,l}P$ where $0 = c_{i,0} < c_{i,1} < c_{i,2} < \dots < c_{i,L_i}$. One may write

$$Z_{i,j} = \sum_{\substack{k=1 \\ k \neq i}}^N \zeta_{k,j} h_{k,i} X_{k,j} + \nu_{i,j} \quad (4)$$

where $\zeta_{k,j}$ is a Bernoulli random variable showing if the k th user has utilized the j th subband, $X_{k,j}$ is the signal of the k th user sent on the j th subband (assuming it has utilized the j th subband), and $\nu_{i,j}$ is the ambient noise which is a zero-mean Gaussian random variable with variance σ^2 . The ratio $\frac{P}{\sigma^2}$ is taken as a measure of SNR and is denoted by γ throughout the paper.

III. ANALYSIS OF THE ACHIEVABLE RATE

Let us denote the achievable rate of the i th user by $\mathcal{R}_i$. It can be observed that the communication channel of this user is a channel with state S_i , the hopping pattern of the i th user, which is independently changing over different transmission slots, and

is known to both the transmitter and the receiver. The achievable rate of such a channel is given by

$$\mathcal{R}_i = I(\vec{X}_i; \vec{Y}_i | S_i) = \sum_{s_i \in \mathcal{S}_i} \Pr(S_i = s_i) I(\vec{X}_i; \vec{Y}_i | S_i = s_i) \quad (5)$$

where $I(\vec{X}_i; \vec{Y}_i | S_i = s_i)$ is the mutual information between $\vec{X}_i$ and $\vec{Y}_i$ for the specific subband selection corresponding to $S_i = s_i$. The set $\mathcal{S}_i$ denotes all possible selections of v_i out of the u subbands. As $p_{\vec{Z}_i}(\cdot)$ is a symmetric density function, meaning all its components have the same pdf given in (3), we deduce that $I(\vec{X}_i; \vec{Y}_i | S_i = s_i)$ is independent of s_i . Therefore, to calculate $\mathcal{R}_i$, we may assume any specific subband selection for the i th user in $\mathcal{S}_i$, say the first v_i subbands. Denoting this specific state by s_i^* , we get

$$\mathcal{R}_i = I(\vec{X}_i; \vec{Y}_i | S_i = s_i^*). \quad (6)$$

In this case, we denote $\vec{Y}_i$ and $\vec{X}_i$ by $\vec{Y}_i(s_i^*)$ and $\vec{X}_i(s_i^*)$, respectively. Obviously, we have

$$\mathcal{R}_i = I(\vec{X}_i(s_i^*); \vec{Y}_i(s_i^*)) = h(\vec{Y}_i(s_i^*)) - h(\vec{Z}_i). \quad (7)$$

Because $\vec{Y}_i(s_i^*)$ and $\vec{Z}_i$ have mixed Gaussian distributions, there is no closed expression for the differential entropy of these vectors. As such, we provide an upper bound and a lower bound on the achievable rate of each user in Sections III-A and III-B and show that these bounds coincide in the asymptotically high SNR regime.

A. Upper Bound on the Achievable Rates

In this section, we develop an upper bound $\mathcal{R}_{i,\text{ub}}$ on the achievable rate of the i th user that is tight enough to ensure that $\mathcal{R}_{i,\text{ub}} - \mathcal{R}_i$ does not increase unboundedly as SNR increases. The idea behind this upper bound is the convexity of $\mathcal{R}_i$ in terms of $p_{\vec{Y}_i(s_i^*)|\vec{X}_i(s_i^*)}(\cdot|\cdot)$.

Theorem 1: There exists an upper bound on the achievable rate of the i th user given by

$$\mathcal{R}_{i,\text{ub}} = \frac{1}{2} v_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{v_k}{u}\right) \log \left(1 + \frac{|h_{i,i}|^2 \gamma}{v_i}\right) + \tilde{\mathcal{R}}_{i,\text{ub}} \quad (8)$$

where $\lim_{\gamma \rightarrow \infty} \tilde{\mathcal{R}}_{i,\text{ub}} < \infty$, i.e.

$$\mathcal{R}_{i,\text{ub}} \sim \frac{1}{2} v_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{v_k}{u}\right) \log \gamma. \quad (9)$$

Proof: Let $\vec{W}_i$ be the $u \times 1$ interference vector where its j th component $W_{i,j}$ is a random variable showing the interference term on the j th frequency subband at the receiver of the i th user. We have $W_{i,j} = \sum_{\substack{k=1 \\ k \neq i}}^N \zeta_{k,j} h_{k,i} X_{k,j}$. Clearly, $\vec{W}_i$ is a mixed Gaussian random vector where the Gaussian components in its pdf represent different choices the other users make in selecting their subbands. In fact, we have $p_{\vec{W}_i}(\vec{w}) = \frac{1}{M_i} \sum_{m=1}^{M_i} g_u(\vec{w}, D_{i,m})$, where

$M_i = \prod_{k \neq i} (v_k)$ and $D_{i,m} = \text{diag}(d_{i,m}^{(1)}, \dots, d_{i,m}^{(u)})$, in which $d_{i,m}^{(j)} = \sum_{k=1}^N \zeta_{k,j,m}^2 |h_{k,i}|^2 \frac{P}{v_k}$ denotes the variance of $W_{i,j}$ for the m th realization of $\{\zeta_{k,j}\}_{k \neq i}$ out of M_i possible realizations that is denoted by $\{\zeta_{k,j,m}\}_{k \neq i}$. If the probability density function of the interference vector consisted only of $g_u(\vec{w}, D_{i,m})$, the forward link of the i th user would be converted into an additive Gaussian channel. The achievable rate of such a virtual channel is given by

$$\begin{aligned} \mathcal{R}_{i,m} &= \frac{1}{2} \log \frac{\det(\text{Cov}(\vec{X}_i(s_i^*)) + D_{i,m} + \sigma^2 I_u)}{\det(D_{i,m} + \sigma^2 I_u)} \\ &= \frac{1}{2} \log \frac{\prod_{j=1}^{v_i} \left(\frac{|h_{i,i}|^2 P}{v_i} + d_{i,m}^{(j)} + \sigma^2 \right)}{\prod_{j=1}^{v_i} (d_{i,m}^{(j)} + \sigma^2)} \\ &= \frac{1}{2} \sum_{j=1}^{v_i} \log \left(1 + \frac{|h_{i,i}|^2 P}{v_i (d_{i,m}^{(j)} + \sigma^2)} \right). \end{aligned} \quad (10)$$

One may also state this as follows. Let

$$T_{i,m} \triangleq \{j : 1 \leq j \leq v_i, d_{i,m}^{(j)} = 0\}. \quad (11)$$

Then

$$\mathcal{R}_{i,m} = \frac{|T_{i,m}|}{2} \log \left(1 + \frac{|h_{i,i}|^2 \gamma}{v_i} \right) + \tilde{\mathcal{R}}_{i,m} \quad (12)$$

where

$$\tilde{\mathcal{R}}_{i,m} = \frac{1}{2} \sum_{1 \leq j \leq v_i: d_{i,m}^{(j)} \neq 0} \log \left(1 + \frac{|h_{i,i}|^2 P}{v_i (d_{i,m}^{(j)} + \sigma^2)} \right) \quad (13)$$

and $|T_{i,m}|$ denotes the cardinality of the set $T_{i,m}$. As each nonzero $d_{i,m}^{(j)}$ is proportional to P , it is clear that $\lim_{\gamma \rightarrow \infty} \tilde{\mathcal{R}}_{i,m} < \infty$. We know that $\mathcal{R}_i$ is convex in terms of $p_{\vec{Y}_i(s_i^*)|\vec{X}_i(s_i^*)}(\vec{y}|\vec{x}) = p_{\vec{Z}_i}(\vec{y} - h_{i,i}\vec{x})$ [60]. Noting this and the fact that $p_{\vec{Z}_i}(\vec{z}) = \frac{1}{M_i} \sum_{m=1}^{M_i} g_u(\vec{z}, D_{i,m} + \sigma^2 I_u)$

$$\begin{aligned} \mathcal{R}_i &\leq \frac{1}{M_i} \sum_{m=1}^{M_i} \mathcal{R}_{i,m} \\ &= \left(\frac{1}{M_i} \sum_{m=1}^{M_i} |T_{i,m}| \right) \frac{1}{2} \log \left(1 + \frac{|h_{i,i}|^2 \gamma}{v_i} \right) \\ &\quad + \tilde{\mathcal{R}}_{i,\text{ub}} \end{aligned} \quad (14)$$

where $\tilde{\mathcal{R}}_{i,\text{ub}} = \frac{1}{M_i} \sum_{m=1}^{M_i} \tilde{\mathcal{R}}_{i,m}$. As each $\tilde{\mathcal{R}}_{i,m}$ saturates by increasing γ , one has $\lim_{\gamma \rightarrow \infty} \tilde{\mathcal{R}}_{i,\text{ub}} < \infty$. Intuitively, $\frac{1}{M_i} \sum_{m=1}^{M_i} |T_{i,m}|$ is the average size of the set of frequencies where the i th user is transmitting and all other users are not transmitting.⁵ Since that the users choose the subbands independently, the probability of only the i th user choosing a particular subband is given by $\frac{v_i}{u} \prod_{k \neq i} (1 - \frac{v_k}{u})$. Since there

are a total of u subbands, the average size is $v_i \prod_{k \neq i} (1 - \frac{v_k}{u})$. Using this in (14), the proof of Theorem 1 is complete. ■

B. Lower Bound on the Achievable Rates

In this section, we derive a lower bound on the achievable rates of users. The idea behind deriving this lower bound is to invoke the classical Entropy Power Inequality (EPI). As we will see, this initial lower bound is not in a closed form as it depends on the differential entropy of a mixed Gaussian random variable. In Appendix A, we obtain an appropriate upper bound on such an entropy that leads us to the final lower bound on $\mathcal{R}_i$.

Theorem 2: There exists a lower bound $\mathcal{R}_{i,\text{lb}}$ on the achievable rate of the i th user which can be written as

$$\mathcal{R}_{i,\text{lb}} = \frac{1}{2} v_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{v_k}{u} \right) \log \gamma + \tilde{\mathcal{R}}_{i,\text{lb}} \quad (15)$$

such that $\lim_{\gamma \rightarrow \infty} \tilde{\mathcal{R}}_{i,\text{lb}} < \infty$, i.e.

$$\mathcal{R}_{i,\text{lb}} \sim \frac{1}{2} v_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{v_k}{u} \right) \log \gamma. \quad (16)$$

Proof: We define $\vec{X}'_i$ to be the $v_i \times 1$ signal vector corresponding to the first v_i elements of $\vec{X}_i(s_i^*)$. Clearly, $\vec{X}'_i$ is a Gaussian vector with covariance matrix $\frac{P}{v_i} I_{v_i}$. Let $\vec{Y}'_i = h_{i,i} \vec{X}'_i + \vec{Z}'_i$ where $\vec{Z}'_i$ is the noise plus interference vector at the receiver side of the i th user on the first v_i subbands. Using EPI

$$2^{\frac{2}{v_i} h(\vec{Y}'_i)} \geq 2^{\frac{2}{v_i} h(h_{i,i} \vec{X}'_i)} + 2^{\frac{2}{v_i} h(\vec{Z}'_i)}. \quad (17)$$

Dividing both sides by $2^{\frac{2}{v_i} h(\vec{Z}'_i)}$, we get

$$h(\vec{Y}'_i) - h(\vec{Z}'_i) \geq \frac{v_i}{2} \log \left(2^{\frac{2}{v_i} (h(h_{i,i} \vec{X}'_i) - h(\vec{Z}'_i))} + 1 \right). \quad (18)$$

On the other hand, since $\vec{Y}'_i$ is a subvector of $\vec{Y}_i(s_i^*)$, we have

$$\begin{aligned} \mathcal{R}_i &= I(\vec{X}_i(s_i^*); \vec{Y}_i(s_i^*)) \geq I(\vec{X}'_i; \vec{Y}'_i) \\ &= h(\vec{Y}'_i) - h(\vec{Z}'_i). \end{aligned} \quad (19)$$

Comparing (18) and (19) yields

$$\mathcal{R}_i \geq \frac{v_i}{2} \log \left(2^{\frac{2}{v_i} (h(h_{i,i} \vec{X}'_i) - h(\vec{Z}'_i))} + 1 \right). \quad (20)$$

Clearly, $h(h_{i,i} \vec{X}'_i) = \frac{v_i}{2} \log \left(2\pi e \frac{|h_{i,i}|^2 P}{v_i} \right)$. As $\vec{Z}'_i$ is a mixed Gaussian random vector, there is no closed formula for $h(\vec{Z}'_i)$. Hence, we have to find an appropriate upper bound on $h(\vec{Z}'_i)$ to further simplify (20). Using the chain rule for the differential entropy

$$h(\vec{Z}'_i) \leq \sum_{j=1}^{v_i} h(Z_{i,j}). \quad (21)$$

Recalling the definitions of $\{a_{i,l}\}_{l=0}^{L_i}$ and $\{c_{i,l}\}_{l=0}^{L_i}$ in the system model, the following Lemma yields an upper bound on $h(Z_{i,j})$ for each $1 \leq j \leq v_i$.

⁴Note that as each user transmits independent Gaussian signals over its chosen subbands, the matrices $\{D_{i,m}\}_{m=1}^{M_i}$ are diagonal.

⁵For a formal proof, see [58].

Lemma 1: For every $1 \leq j \leq v_i$ and for all values of γ , there exists an upper bound on $h(Z_{i,j})$ given by

$$h(Z_{i,j}) \leq \frac{1 - a_{i,0}}{2} \log(c_{i,L_i} \gamma + 1) + \log(\sqrt{2\pi e} \sigma) - \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l}. \quad (22)$$

Proof: In Appendix A, we derive an upper bound on the differential entropy of a complex mixed Gaussian random variable in (130). Modifying the result in Appendix A appropriately for the real mixed Gaussian random variable $Z_{i,j}$, we have

$$h(Z_{i,j}) \leq \frac{1}{2} \sum_{l=0}^{L_i} a_{i,l} \log(2\pi e \sigma_{i,l}^2) - \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l}. \quad (23)$$

Noting that $\sigma_{i,0} = \sigma$ and $\sigma_{i,l}^2 \leq \sigma^2(c_{i,L_i} \gamma + 1)$ for any $1 \leq l \leq L_i$

$$h(Z_{i,j}) \leq \frac{1 - a_{i,0}}{2} \log(c_{i,L_i} \gamma + 1) + \log(\sqrt{2\pi e} \sigma) - \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l}. \quad (24)$$

■

By (20), (21), and (22)

$$\begin{aligned} \mathcal{R}_i &\geq \mathcal{R}_{i,\text{lb}} \triangleq \frac{v_i}{2} \log \left(\frac{2^2 \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l} |h_{i,i}|^2 \gamma}{v_i (c_{i,L_i} \gamma + 1)^{1-a_{i,0}}} + 1 \right) \\ &= \frac{v_i}{2} \log \left(\frac{2^2 \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l} |h_{i,i}|^2}{v_i (c_{i,L_i} + \gamma^{-1})^{1-a_{i,0}}} + \gamma^{-a_{i,0}} \right) \\ &\quad + \frac{1}{2} v_i a_{i,0} \log \gamma. \end{aligned} \quad (25)$$

Defining $\tilde{\mathcal{R}}_{i,\text{lb}} \triangleq \frac{v_i}{2} \log \left(\frac{2^2 \sum_{l=0}^{L_i} a_{i,l} \log a_{i,l} |h_{i,i}|^2}{v_i (c_{i,L_i} + \gamma^{-1})^{1-a_{i,0}}} + \gamma^{-a_{i,0}} \right)$,

we note that $\lim_{\gamma \rightarrow \infty} \tilde{\mathcal{R}}_{i,\text{lb}} < \infty$. Combining this with the fact that $a_{i,0} = \prod_{k=1}^N \left(1 - \frac{v_k}{u}\right)$, the proof of Theorem 2 is complete.⁶ ■

One may consider the following generalization of the FH scheme. Let us assume that the users are not restricted to choose a fixed number of frequency subbands in each transmission slot. In fact, in each transmission slot the number of selected subbands can be any integer between 0 and u , and the probability of choosing $v \in [0, u] \cap \mathbb{Z}$ subbands by the i th user is denoted by $\mu_{i,v}$. Therefore, the i th user has two random generators. The first random generator selects a number $v \in [0, u] \cap \mathbb{Z}$ according to the probability mass function $\{\mu_{i,v}\}_{v=0}^u$, while the other generator selects v subbands among the whole available u subbands.

⁶In [58], we address another approach to propose a lower bound (looser than $\mathcal{R}_{i,\text{lb}}$) on the achievable rate of the i th user with the same SNR scaling as $\mathcal{R}_{i,\text{lb}}$.

This repeats independently from transmission slot to transmission slot. Based on the arguments made at the beginning of this section, the achievable rate of the i th user can be written as

$$\mathcal{R}_i = \sum_{v=0}^u \mu_{i,v} I(\vec{X}_i(s_{i,v}^*); \vec{Y}_i(s_{i,v}^*)) \quad (26)$$

where $s_{i,v}^*$ denotes the state where the i th user selects the first v subbands. Clearly, $I(\vec{X}_i(s_{i,0}^*); \vec{Y}_i(s_{i,0}^*)) = 0$ for any $1 \leq i \leq N$. Furthermore

$$\begin{aligned} a_{i,0} &= \Pr\{Z_{i,1} \text{ is interference-free}\} \\ &= \sum_{v_1=0}^u \cdots \sum_{v_{i-1}=0}^u \sum_{v_{i+1}=0}^u \cdots \sum_{v_N=0}^u \prod_{k=1, k \neq i}^N \mu_{k,v_k} \left(1 - \frac{v_k}{u}\right) \\ &= \prod_{k=1, k \neq i}^N \sum_{v_k=0}^u \mu_{k,v_k} \left(1 - \frac{v_k}{u}\right) \\ &= \prod_{k=1, k \neq i}^N \sum_{v=0}^u \mu_{k,v} \left(1 - \frac{v}{u}\right) \\ &= \prod_{k=1, k \neq i}^N \left(1 - \frac{\bar{v}_k}{u}\right) \end{aligned} \quad (27)$$

where $\bar{v}_k \triangleq \sum_{v=0}^u v \mu_{k,v}$. Applying the results of this section, we get

$$I(\vec{X}_i(s_{i,v}^*); \vec{Y}_i(s_{i,v}^*)) \sim \frac{1}{2} v a_{i,0} \log \gamma. \quad (28)$$

By (26), (27), and (28)

$$\begin{aligned} \mathcal{R}_i &\sim \sum_{v=0}^u \frac{1}{2} \mu_{i,v} v \prod_{k=1, k \neq i}^N \left(1 - \frac{\bar{v}_k}{u}\right) \log \gamma \\ &= \frac{1}{2} \bar{v}_i \prod_{k=1, k \neq i}^N \left(1 - \frac{\bar{v}_k}{u}\right) \log \gamma. \end{aligned} \quad (29)$$

In fact, (29) demonstrates that the generalized FH scheme is equivalent to the FH scheme through substituting $\{v_i\}_{i=1}^N$ by $\{\bar{v}_i\}_{i=1}^N$. However, it is remarkable that in contrast with the FH scheme in which $\{v_i\}_{i=1}^N$ are integer values, in the generalized FH scheme $\{\bar{v}_i\}_{i=1}^N$ are real values. This provides more flexibility in system design. The above observation motivates us to use this generalized scenario in the rest of the paper and we simply refer to it as the FH scheme. In this scheme, the i th user has a parameter $\bar{v}_i$, which can be chosen to be any real number in the interval $[0, u]$.

IV. SYSTEM DESIGN

In this section, we find the optimum operation point for the FH scheme. This requires finding the optimum values of $\{\bar{v}_i\}_{i=1}^N$. Based on the results established in the previous section, there exist upper and lower bounds on the achievable

rate of each user that coincide in the high SNR regime. As such, the achievable rate itself must be asymptotically equivalent to each of these bounds, i.e.

$$\mathcal{R}_i \sim \frac{1}{2} \bar{v}_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{\bar{v}_k}{u}\right) \log \gamma \quad (30)$$

where, based on the conclusion made at the end of Section III, the parameters $\{\bar{v}_i\}_{i=1}^N$ can be adjusted to be any real number in the range $[0, u]$. By (30), the network sum rate can be asymptotically written as

$$\sum_{i=1}^N \mathcal{R}_i \sim \text{SMG}(\bar{v}_1, \dots, \bar{v}_N) \log \gamma \quad (31)$$

where

$$\text{SMG}(\bar{v}_1, \dots, \bar{v}_N) \triangleq \sum_{i=1}^N \frac{1}{2} \bar{v}_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{\bar{v}_k}{u}\right). \quad (32)$$

We call $\text{SMG}(\bar{v}_1, \dots, \bar{v}_N)$ the sum multiplexing gain of the system. $\text{SMG}(\bar{v}_1, \dots, \bar{v}_N)$ is a symmetric function of $(\bar{v}_1, \dots, \bar{v}_N)$ and has a saddle point at $\bar{v}_i = \frac{u}{N}$ for $1 \leq i \leq N$. In a fair FH system, it is required that $\bar{v}_i = v$ for all $1 \leq i \leq N$ where v is any real number in the interval $[0, u]$. We call v the hopping parameter in the randomized FH scenario. Hence, we define

$$\begin{aligned} \text{SMG}(v, N) &\triangleq \text{SMG}(\bar{v}_1, \dots, \bar{v}_N) \big|_{\forall i: \bar{v}_i = v} \\ &= \frac{N}{2} v \left(1 - \frac{v}{u}\right)^{N-1}. \end{aligned} \quad (33)$$

Maximizing this in terms of v yields⁷

$$v_{\text{opt}} = \frac{u}{N}. \quad (34)$$

Setting $v = v_{\text{opt}}$, the highest sum multiplexing gain of the fair FH scheme is given by

$$\sup_v \text{SMG}(v, N) = \frac{1}{2} u \left(1 - \frac{1}{N}\right)^{N-1}. \quad (35)$$

It is remarkable that $\frac{u}{N}$ may not be an integer. If we do not adopt the generalized FH scheme, then all users must hop randomly over sets of $\tilde{v} = \max\{\lfloor \frac{u}{N} \rfloor, 1\}$ frequency subbands. This results in a sum multiplexing gain of $\frac{N}{2} \tilde{v} \left(1 - \frac{\tilde{v}}{u}\right)^{N-1}$. This is generally less than $\frac{1}{2} u \left(1 - \frac{1}{N}\right)^{N-1}$. By adopting the generalized FH scheme in case $\frac{u}{N} \notin \mathbb{Z}$, each user only needs to hop randomly over different sets of frequency subbands of cardinality $\lfloor \frac{u}{N} \rfloor$ or $\lceil \frac{u}{N} \rceil$. In fact, each user has two random generators. The first random generator selects one of the numbers $\lfloor \frac{u}{N} \rfloor$ and $\lceil \frac{u}{N} \rceil$ with probabilities μ and $1 - \mu$, respectively, such that $\mu \lfloor \frac{u}{N} \rfloor + (1 - \mu) \lceil \frac{u}{N} \rceil = \frac{u}{N}$ or equivalently $\mu = \lceil \frac{u}{N} \rceil - \frac{u}{N}$. Let us assume the first random generator has selected a number

$a \in \{\lfloor \frac{u}{N} \rfloor, \lceil \frac{u}{N} \rceil\}$. Then, the second random generator selects a subset of cardinality a among the u frequency subbands. Doing this independently from transmission slot to transmission slot, the sum multiplexing gain given in (35) is achieved.

Observation 1: One might suggest another well-known utility function that is popular in the game theory context, namely the proportional fair function, which is defined as $\sum_{i=1}^N \log \mathcal{R}_i$. We have

$$\begin{aligned} \sum_{i=1}^n \log \mathcal{R}_i &\sim \sum_{i=1}^N \log \left(\frac{1}{2} \bar{v}_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{\bar{v}_k}{u}\right) \log \gamma \right) \\ &= \sum_{i=1}^N \log \left(\frac{1}{2} \bar{v}_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{\bar{v}_k}{u}\right) \right) \\ &\quad + N \log \log \gamma. \end{aligned} \quad (36)$$

It can be easily verified that $\sum_{i=1}^N \log \left(\frac{1}{2} \bar{v}_i \prod_{\substack{k=1 \\ k \neq i}}^N \left(1 - \frac{\bar{v}_k}{u}\right) \right)$ has an absolute maximum at $\bar{v}_i = \frac{u}{N}$ for $1 \leq i \leq N$.

Observation 2: As we will discuss in more detail in Section V, the number of active users in the system is in general a random variable $\mathfrak{N}$ with realization N . Although users in the FH system can use their knowledge about the number of active users to adjust the hopping parameter (as explained earlier in this section), one may devise a suboptimal rule to fix $v = v^*$ given by

$$v^* = \arg \max_{v \in [0, u]} E\{\text{SMG}(v, \mathfrak{N})\} \quad (37)$$

where the expectation is with respect to the number of active users in the network. This selection of v by all users makes the system robust against changes in the number of active users in the network.⁸ We call this version of the FH system the robust randomized FH. We remark that the rule in (37) is a particular design approach for the robust FH system. In Section V, we consider another design rule based on maximizing the average of the minimum multiplexing gain per user in term of the number of active users in the network.

V. COMPARISON OF THE ROBUST FH SCENARIO WITH THE FD SCHEME

In a centralized setup, under the condition that no user is aware of the other users' codebooks and the number of users is fixed and known to the central controller, it is shown in [9] that if the crossover channel gains are sufficiently larger than the forward channel gains, then every Pareto optimal rate vector is realized by FD for all ranges of SNR. However, in realistic scenarios, the number of active users is not fixed. This degrades the performance of the FD scheme as it is designed for a specified number of users. In particular, if the number of active users is less than the designed target of the FD scheme, a considerable portion of the spectrum may remain unused. This encourages us

⁷Computation of v_{opt} requires that all transmitters know the number of active users N in the network. This is more discussed in Section V.

⁸In fact, the transmitters use their knowledge about the instantaneous number of active users only to regulate their transmission rates. This is explained in (42).

to compare the performance of the proposed robust FH scheme with that of the FD scheme in a setup where the number of active users is a random variable $\mathfrak{N}$ with a given distribution.⁹ The realization of $\mathfrak{N}$ is denoted by N as before.

To perform the comparison, we introduce two different performance measures. In the following definitions, the sup operation is over possible adjustable parameters in the system, e.g., the hopping parameter in the robust randomized FH scenario. All expectations are taken with respect to $\mathfrak{N}$. We define $q_n \triangleq \Pr\{\mathfrak{N} = n\}$ for all $n \geq 0$. It is assumed that the maximum number of active users in the network is $n_{\max}$, i.e., $\Pr\{\mathfrak{N} > n_{\max}\} = 0$. We usually take $q_0 = 0$ unless otherwise stated.

- **Average Sum Multiplexing Gain**, which is defined as

$$\eta^{(1)} \triangleq \sup \lim_{\gamma \rightarrow \infty} \frac{E\left\{\sum_{i=1}^{\mathfrak{N}} \mathcal{R}_i\right\}}{\log \gamma} = \sup E\{\text{SMG}\} \quad (38)$$

where $\text{SMG} = \lim_{\gamma \rightarrow \infty} \frac{\sum_{i=1}^{\mathfrak{N}} \mathcal{R}_i}{\log \gamma}$ is the sum multiplexing gain.

- **Average Minimum Multiplexing Gain per User**, which is defined as

$$\eta^{(2)} \triangleq \sup \lim_{\gamma \rightarrow \infty} \frac{E\{\min_{1 \leq i \leq \mathfrak{N}} \mathcal{R}_i\}}{\log \gamma}. \quad (39)$$

The FD system is designed to service a certain number of active users. We denote this design target in the FD scheme by n_{des} . Therefore, the spectrum is divided to n_{des} bands where each band contains $\frac{u}{n_{\text{des}}}$ frequency subbands. This requires that u is divisible by n_{des} , which is assumed to be the case to guarantee fairness. Each user that becomes active occupies an empty band. If there is no empty band, no service is available. In case $n_{\max} \leq u$, the central controller in the FD system sets $n_{\text{des}} = n_{\max}$ to ensure that all users can receive service upon activation.¹⁰ In case $n_{\max} > u$, the central controller sets $n_{\text{des}} = u$ to guarantee that as many users receive service as possible. Therefore, $n_{\text{des}} = \min\{n_{\max}, u\}$. In fact, one can show that selecting $n_{\text{des}} = \min\{n_{\max}, u\}$ maximizes the service capability in the FD system. Service capability is measured by $E\left\{\frac{\mathfrak{N}_{\text{serv}}}{\mathfrak{N}}\right\}$ where $\mathfrak{N}_{\text{serv}}$ denotes the number of active users receiving service, i.e., their multiplexing gain is strictly positive.

Due to the nature of the randomized FH scheme, as far as the hopping parameter v is strictly less than u , all users receive service, while if $v = u$ and $N > 1$, no user receives service, i.e., the multiplexing gain achieved by any active user is zero. As such, to get the largest service capability in the FH scenario, we require $v \in (0, u)$. As an example, if v^* in (37) is equal to u , the service capability will be less than 1. To avoid this, we set the hopping parameter $v = v^* - \Delta = u - \Delta$ for sufficiently small Δ such that the performance of the robust FH is still above the performance of the FD scenario.

⁹As explicitly mentioned in the system model, the number of users is assumed to be fixed for the whole transmission period of interest.

¹⁰We assume u is divisible by $n_{\max}$ in this case.

Note that the comparison between the robust FH and the FD schemes is a fair comparison based on the following facts:

- 1 Both schemes are required to achieve the largest possible service capability. This is enforced by setting $v \in (0, u)$ and $n_{\text{des}} = \min\{n_{\max}, u\}$ in the robust FH and FD scenarios, respectively.
- 2 The hopping parameter v in robust FH and the number of bands n_{des} in FD are both designed based on the distribution of the number of active users rather than the realization of the number of active users itself.

Average Sum Multiplexing Gain: This measure is a meaningful tool of comparison if $n_{\max} < \infty$. Hence, we assume $n_{\max}$ is a finite number and u is a multiple of $n_{\max}$ in this subsection. It is easily seen that the sum multiplexing gain in the FD scenario is

$$\text{SMG}_{\text{FD}}(n_{\text{des}}, \mathfrak{N}) = \begin{cases} \frac{\mathfrak{N}}{2} \frac{u}{n_{\text{des}}} & \mathfrak{N} \leq n_{\text{des}} \\ \frac{u}{2} & \mathfrak{N} > n_{\text{des}} \end{cases}. \quad (40)$$

By (33), $\text{SMG}_{\text{FH}}(v, \mathfrak{N})$ is given by

$$\text{SMG}_{\text{FH}}(v, \mathfrak{N}) = \frac{1}{2} \mathfrak{N} v \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1}. \quad (41)$$

As aforementioned, a candidate for robust hopping strategy against changes in the number of active users is the one given in (37). It is notable that although the value of v is fixed at v^* , any user regulates its transmission rate based on the instantaneous number of active users to avoid transmission failure. Using the lower bound on the achievable rate of the i th user given in (25), it is shown in Appendix B that the i th user may select its transmission rate R_i as

$$R_i = E \left\{ \frac{V_i}{2} \log \left(\frac{2^{-2(N-1)} \left(\vartheta_{v^*} + \frac{\mathcal{H}(\lceil v^* \rceil - v^*)}{V_i} \right) \gamma |h_{i,i}|^2}{V_i \left(\frac{\gamma \sum_{j \neq i} |h_{j,i}|^2}{\lfloor v^* \rfloor} + 1 \right)^{1 - \left(1 - \frac{v^*}{u}\right)^{N-1}}} + 1 \right) \right\} \quad (42)$$

where V_i is a random variable taking the values $\lfloor v^* \rfloor$ and $\lceil v^* \rceil$ with probabilities $\lceil v^* \rceil - v^*$ and $v^* - \lfloor v^* \rfloor$, respectively. The quantity ϑ_{v^*} is defined in

$$\vartheta_{v^*} \triangleq \begin{cases} \mathcal{H}\left(\frac{v^*}{u}\right) & v^* \in \mathbb{N} \\ (v^* - \lfloor v^* \rfloor) \mathcal{H}\left(\frac{\lfloor v^* \rfloor}{u}\right) + (\lceil v^* \rceil - v^*) \mathcal{H}\left(\frac{\lceil v^* \rceil}{u}\right) & v^* \notin \mathbb{N} \end{cases}. \quad (43)$$

It is seen that the quantities the i th transmitter needs to evaluate R_i are $|h_{i,i}|$, $\sum_{j \neq i} |h_{j,i}|^2$, and N . The i th receiver sends these required data to its transmitter via a feedback link. It is shown in [58, Appendix B] how the i th receiver can estimate N and $\{h_{j,i}\}_{j \neq i}$ using a combined technique based on the method of moments and maximum likelihood estimation [63]. In this paper, we assume the i th user has perfect knowledge of N and the channel gains $\{h_{j,i}\}_{j=1}^N$. In Section VII, we relax this assumption partially by not requiring any knowledge about the channel gains and the number of active users at the transmitters. Moreover, regulating the transmission rate at R_i by the i th user

for any realization N of $\mathfrak{N}$, one may observe that an average sum multiplexing gain of $E \left\{ \mathfrak{N} v^* \left(1 - \frac{v^*}{u} \right)^{\mathfrak{N}-1} \right\}$ is achievable.

We present an example to compare the performance of FH with that of FD in terms of $\eta^{(1)}$.

Example 1: Let us consider a network where $n_{\max} = 2$. The central controller in the FD system sets $n_{\text{des}} = 2$, and according to (40), $\eta_{\text{FD}}^{(1)} = E\{\text{SMG}_{\text{FD}}(2, \mathfrak{N})\} = q_1 \frac{u}{4} + q_2 \frac{u}{2} = \frac{q_1 + 2q_2}{4} u$. Based on (41), $E\{\text{SMG}_{\text{FH}}(v, \mathfrak{N})\} = \frac{1}{2} q_1 v + q_2 v \left(1 - \frac{v}{u} \right)$. Using this in (37)

$$\begin{aligned} v^* &= \arg \max_{v \in [0, u]} E\{\text{SMG}_{\text{FH}}(v, \mathfrak{N})\} \\ &= \begin{cases} \frac{q_1 + 2q_2}{4} u & q_1 \leq 2q_2 \\ u & q_1 > 2q_2 \end{cases}. \end{aligned} \quad (44)$$

Therefore

$$\begin{aligned} \eta_{\text{FH}}^{(1)} &= \sup_{v \in [0, u]} E\{\text{SMG}_{\text{FH}}(v, \mathfrak{N})\} \\ &= E\{\text{SMG}_{\text{FH}}(v^*, \mathfrak{N})\} \\ &= \begin{cases} \frac{(q_1 + 2q_2)^2}{16q_2} u & q_1 \leq 2q_2 \\ \frac{q_1}{2} u & q_1 > 2q_2 \end{cases}. \end{aligned} \quad (45)$$

It is easy to see that $\eta_{\text{FH}}^{(1)} > \eta_{\text{FD}}^{(1)}$ if and only if $q_1 > 2q_2$, or equivalently, $q_1 > \frac{2}{3}$. We note that in this case $v^* = u$, i.e., all users spread their power on the whole spectrum and no hopping is performed. This makes service capability be strictly less than %100 because, if both users are active, none of them receive service. As such, we take $v = u - \Delta$. To ensure that the performance of the robust FH scenario is above that of the FD system, we require

$$\frac{1}{2} q_1 (u - \Delta) + q_2 (u - \Delta) \left(1 - \frac{u - \Delta}{u} \right) > \frac{q_1 + 2q_2}{4} u. \quad (46)$$

As far as $\Delta < \frac{u}{2}$, (46) is equivalent to $q_1 > 2q_2 \frac{1 - 2\Delta(1 - \frac{\Delta}{u})}{1 - 2\Delta}$. This is a more restrictive condition than $q_1 > 2q_2$ which is the cost paid for having full service capability. $\square$

Average Minimum Multiplexing Gain Per User: This measure can also be written as

$$\eta^{(2)} = \sup E \left\{ \frac{\text{SMG}}{\mathfrak{N}} \mathbf{1}(\mathfrak{N}_{\text{serv}} = \mathfrak{N}) \right\}. \quad (47)$$

In fact, if $\mathfrak{N}_{\text{serv}} \neq \mathfrak{N}$, there exists at least one user that achieves no multiplexing gain. Therefore, the minimum multiplexing gain per user is zero in this case. However, if $\mathfrak{N}_{\text{serv}} = \mathfrak{N}$, all users achieve a nonzero multiplexing gain. This measure can be used whether $n_{\max}$ is finite or infinite.

In case of the FH scenario, the rule to choose the optimum value of the hopping parameter v , denoted by v_* , is given by

$$v_* = \arg \max_{v \in [0, u]} E \left\{ \frac{\text{SMG}_{\text{FH}}(v, \mathfrak{N})}{\mathfrak{N}} \mathbf{1}(\mathfrak{N}_{\text{serv}} = \mathfrak{N}) \right\}. \quad (48)$$

In this case, the transmission rate of the i th user is given by (42) where v^* is replaced by v_* .

In general, it is hard to determine v^* or v_* analytically for $n_{\max} \geq 3$. Therefore, one can not obtain a closed characterization for the regions $\{ \{q_n\}_{n=1}^{n_{\max}} : \eta_{\text{FH}}^{(1)} > \eta_{\text{FD}}^{(1)} \}$ and $\{ \{q_n\}_{n=1}^{n_{\max}} : \eta_{\text{FH}}^{(2)} > \eta_{\text{FD}}^{(2)} \}$. We aim to derive sufficient conditions on $\{q_n\}_{n=1}^{n_{\max}}$ (inner bounds to the mentioned regions) such that $\eta_{\text{FH}}^{(1)} > \eta_{\text{FD}}^{(1)}$ or $\eta_{\text{FH}}^{(2)} > \eta_{\text{FD}}^{(2)}$. Our conditions only involve $E\{\mathfrak{N}\}$ and $n_{\max}$.

Proposition 1: If

$$E\{\mathfrak{N}\} < \frac{1}{2} \ln((e^2 - 1)n_{\max}) \quad (49)$$

then, $\eta_{\text{FD}}^{(1)} < \eta_{\text{FH}}^{(1)}$. Also, as far as

$$\frac{1}{E\{\mathfrak{N}\}} \left(1 - \frac{1}{E\{\mathfrak{N}\}} \right)^{E\{\mathfrak{N}\}-1} > \frac{1}{n_{\max}} \quad (50)$$

we have $\eta_{\text{FD}}^{(2)} < \eta_{\text{FH}}^{(2)}$.

Proof: See Appendix C. $\blacksquare$

We consider an example where $n_{\max}$ is not finite.

Example 2: We assume a Poisson distribution on the number of active users, i.e., $q_n = \frac{e^{-\lambda} \lambda^n}{n!}$ for $n \geq 0$ and $\lambda > 0$. We have

$$\begin{aligned} &E \left\{ \frac{\text{SMG}_{\text{FH}}(v, \mathfrak{N})}{\mathfrak{N}} \mathbf{1}(\mathfrak{N}_{\text{serv, FH}} = \mathfrak{N}) \right\} \\ &\stackrel{(a)}{=} E \left\{ \frac{\text{SMG}_{\text{FH}}(v, \mathfrak{N})}{\mathfrak{N}} \right\} \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \left(v \left(1 - \frac{v}{u} \right)^{n-1} \right) \\ &= \frac{1}{2} \frac{v}{1 - \frac{v}{u}} \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \left(1 - \frac{v}{u} \right)^n \\ &\stackrel{(b)}{=} \frac{1}{2} \frac{v}{1 - \frac{v}{u}} \left(e^{\lambda \left(e^{\ln(1 - \frac{v}{u})} - 1 \right)} - e^{-\lambda} \right) \\ &= \frac{e^{-\lambda} (1 - \omega_v) (e^{\lambda \omega_v} - 1)}{2 \omega_v} u. \end{aligned} \quad (51)$$

In the above equation, (a) results from the fact that $\mathbf{1}(\mathfrak{N}_{\text{serv, FH}} = \mathfrak{N}) = 0$ whenever $v = u$ and $N > 1$, however, $\text{SMG}_{\text{FH}}(v, \mathfrak{N}) = 0$ in this case. Also, (b) follows by the fact that $E\{e^{t\mathfrak{N}}\} = e^{\lambda(e^t - 1)}$ for any $t \in \mathbb{R}$ and we have defined $\omega_v \triangleq 1 - \frac{v}{u}$. It can be easily seen that the optimal $v = v_*$ satisfies the nonlinear equation $e^{-\lambda \omega_{v_*}} = 1 - \lambda \omega_{v_*} (1 - \omega_{v_*})$. Solving this for v_* , we find out that v_* is not equal to u for all $\lambda > 2$. The following table lists the values of ω_{v_*} , the values of v_* and also the corresponding average minimum multiplexing gain per user $\eta_{\text{FH}}^{(2)}$ for $\lambda \in \{3, 4, 5, 6\}$.

λ	3	4	5	6
ω_{v_*}	0.4536	0.6392	0.7347	0.7912
v_*	0.5464u	0.3608u	0.2653u	0.2088u
$\eta_{\text{FH}}^{(2)}$	0.0869u	0.0615u	0.0467u	0.0374u

(52)

The FD system aims to serve as many users as it can. Since it is not possible to serve more than u users, the number of bands is

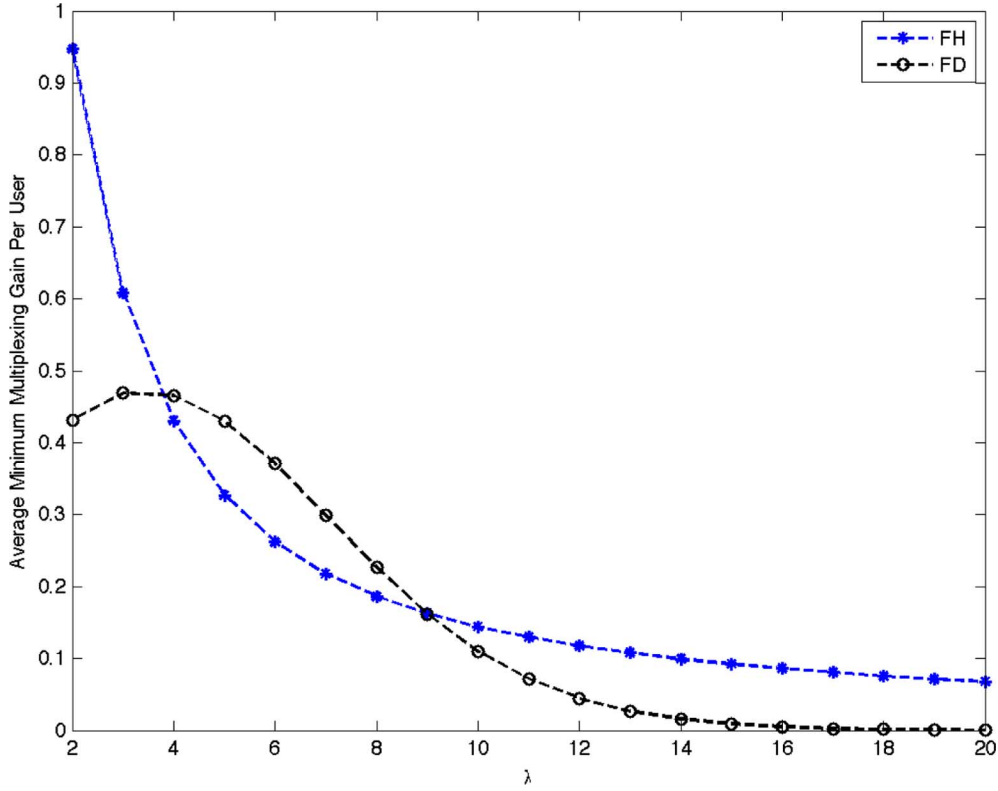


Fig. 1. Curves of $\eta_{\text{FH}}^{(2)}$ and $\eta_{\text{FD}}^{(2)}$ in terms of λ in a network with $u = 7$ subbands.

set at $n_{\text{des}} = u$. Therefore, $\mathfrak{N}_{\text{serv,FD}} < \mathfrak{N}$ if and only if $\mathfrak{N} > u$. Using this and by (40)

$$\begin{aligned} \eta_{\text{FD}}^{(2)} &= E \left\{ \frac{\text{SMG}_{\text{FD}}(n_{\text{des}}, \mathfrak{N})}{\mathfrak{N}} \mathbf{1}(\mathfrak{N}_{\text{serv,FD}} = \mathfrak{N}) \right\} \\ &= E \left\{ \frac{\text{SMG}_{\text{FD}}(u, \mathfrak{N})}{\mathfrak{N}} \middle| \mathfrak{N} \leq u \right\} \Pr\{\mathfrak{N} \leq u\} \\ &= \frac{1}{2} \sum_{n=1}^u \frac{e^{-\lambda} \lambda^n}{n!}. \end{aligned} \quad (53)$$

We have sketched $\eta_{\text{FH}}^{(2)}$ and $\eta_{\text{FD}}^{(2)}$ in terms of λ in Fig. 1 for $u = 7$. It is noticeable that $\eta_{\text{FH}}^{(2)}$ scales linearly with u . However, $\eta_{\text{FD}}^{(2)}$ is always less than $\frac{1}{2}$ no matter how large u is. As u increases, the advantage of FH over FD becomes more apparent. $\square$

VI. ADAPTIVE FREQUENCY HOPPING (AFH)

The results of the previous section are obtained based on the assumption that the hopping parameter v is fixed and is not adaptively changed based on the number of active users. The performance of the FH system can be improved by letting the transmitters adapt their hopping parameter based on the number of active users using (34). We refer to this scenario as AFH. In the following example, we study the performance improvement offered by AFH over FH in terms of $\eta^{(1)}$ and $\eta^{(2)}$.

Example 3: Let us assume that the number of active users is a Poisson random variable with parameter $\lambda > 1$. To compute v^* in the FH scenario, we have

$$\begin{aligned} E\{\text{SMG}_{\text{FH}}(v, \mathfrak{N})\} &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \left(nv \left(1 - \frac{v}{u}\right)^{n-1} \right) \\ &= \frac{1}{2} v \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n-1)!} \left(1 - \frac{v}{u}\right)^{n-1} \\ &= \frac{1}{2} \lambda v \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \left(1 - \frac{v}{u}\right)^n \end{aligned} \quad (54)$$

$$= \frac{1}{2} \lambda v e^{-\frac{\lambda v}{u}}. \quad (55)$$

Therefore

$$v^* = \arg \max_v E\{\text{SMG}_{\text{FH}}(v, \mathfrak{N})\} = \frac{u}{\lambda}. \quad (56)$$

Since $\lambda \neq 1$, we get $v^* \neq u$. Thus, choosing $v = v^*$ guarantees a 100% service capability. Finally

$$\eta_{\text{FH}}^{(1)} = \frac{u}{2e}. \quad (57)$$

By (35)

$$\eta_{\text{AFH}}^{(1)} = \frac{u}{2} \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \left(1 - \frac{1}{n}\right)^{n-1}. \quad (58)$$

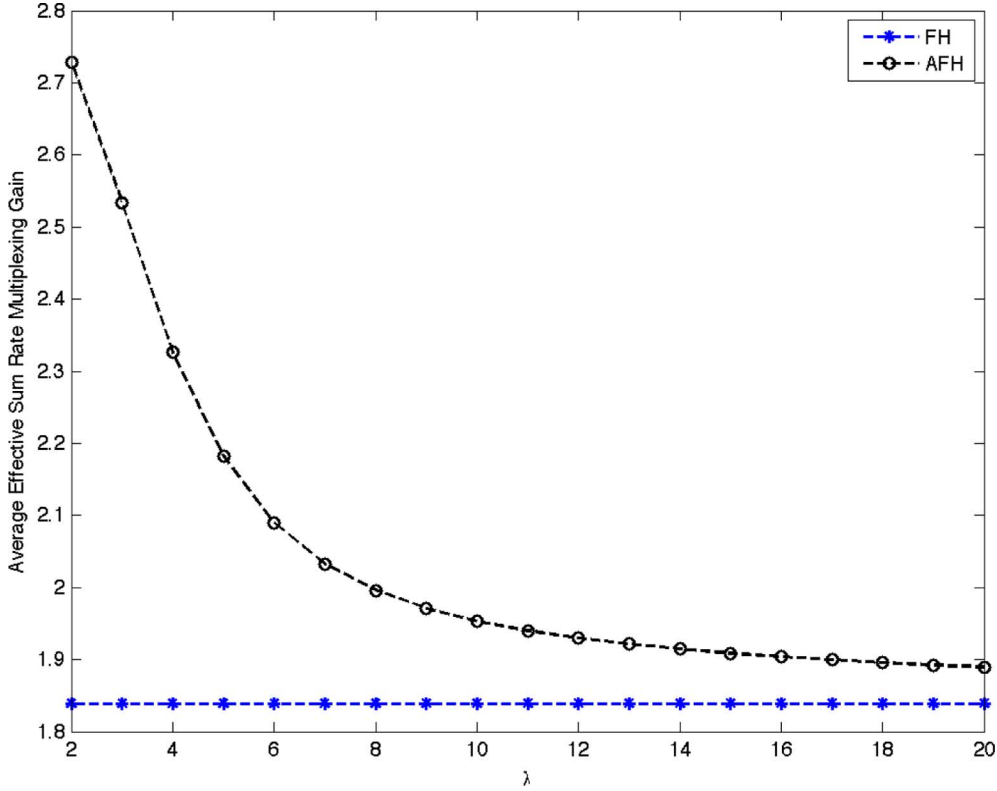


Fig. 2. Plots of $\eta_{\text{AFH}}^{(1)}$ and $\eta_{\text{FH}}^{(1)}$ in terms of λ for $u = 10$.

Fig. 2 shows the plots of $\eta_{\text{FH}}^{(1)}$ and $\eta_{\text{AFH}}^{(1)}$ versus λ for $u = 10$. It is observed that $\eta_{\text{FH}}^{(1)}$ does not change with λ , while $\eta_{\text{AFH}}^{(1)}$ decreases by increasing λ . This indicates that in a crowded network (large λ), AFH does not provide any significant advantage over FH in terms of $\eta^{(1)}$.

We have already calculated $\eta_{\text{FH}}^{(2)}$ in example 2 in a system where $3 \leq \lambda \leq 10$. Also

$$\eta_{\text{AFH}}^{(2)} = \frac{u}{2} \sum_{n=1}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}. \quad (59)$$

Fig. 3 presents the plots of $\eta_{\text{FH}}^{(2)}$ and $\eta_{\text{AFH}}^{(2)}$ versus λ for $u = 10$. Both $\eta_{\text{FH}}^{(2)}$ and $\eta_{\text{AFH}}^{(2)}$ decrease by increasing λ . However, the ratio $\frac{\eta_{\text{AFH}}^{(2)}}{\eta_{\text{FH}}^{(2)}}$ decreases as λ increases. This indicates that for large values of λ , AFH does also not provide any significant advantage over FH in terms of $\eta^{(2)}$. $\square$

VII. ANALYSIS OF THE OUTAGE CAPACITY

As shown in Section V, the transmitter of the i th user requires N , $|h_{i,i}|$, and $\sum_{j \neq i} |h_{j,i}|^2$ to regulate its transmission rate. Assuming the transmitters are not aware of the channel gains and the number of active users in the network, the Shannon capacity is not meaningful anymore. In this case, a suitable performance measure is the ε -outage capacity, denoted by $R(\varepsilon)$, which is defined as the maximum transmission rate per user ensuring an outage probability below ε . Throughout the rest of the paper, N is considered to be a realization of the random variable $\mathfrak{N}$. Also, we assume the channel gains $\{h_{i,j}\}_{i,j=1}^N$ are realizations of independent and circularly symmetric zero-mean complex Gaussian random variables

with unit variances denoted by $\{h_{i,j}\}_{i,j=1}^{\mathfrak{N}}$. Accordingly, $c_{i,l}$ and $a_{i,l}$ are realizations of the random variables $\mathbf{c}_{i,l}$ and $\mathbf{a}_{i,l}$ for $1 \leq i \leq \mathfrak{N}$ and $0 \leq l \leq 2^{\mathfrak{N}-1} - 1$. Since the channel gains are complex, we consider the randomized FH scheme in the complex setup where users transmit independent and circularly symmetric complex Gaussian signals. This causes minor changes in the formulation of the bounds developed earlier. For tractability reasons, we only deal with the primary version of randomized FH, i.e., all users only hop over a *fixed number* $v_1 = v_2 = \dots = v_{\mathfrak{N}} = v$ of frequency subbands from transmission slot to transmission slot where $0 \leq v \leq u$ is an integer. Moreover, $L_i = 2^{N-1} - 1$ is a realization of $\mathcal{L}_i = 2^{\mathfrak{N}-1} - 1$.

Let $\vec{h}_i$ contain the channel coefficients concerning the i th user, i.e., $\vec{h}_i = (h_{1,i} \dots h_{\mathfrak{N},i})^T$. In this case, we denote the achievable rate of the i th user by $\mathcal{R}_i(\vec{h}_i)$. The outage event for this user is

$$\mathcal{O}_i(R) \triangleq \{\vec{h}_i : \mathcal{R}_i(\vec{h}_i) < R\} \quad (60)$$

where R is the transmission rate. Hence

$$R(\varepsilon) = \sup\{R : \Pr\{\mathcal{O}_i(R)\} < \varepsilon\}. \quad (61)$$

We emphasize that the randomness of the number of active users is involved in the outage event, as $\mathfrak{N}$ represents the size of $\vec{h}_i$. Moreover, due to symmetry, the ε -outage capacity is the same for all users.

Since there is no closed formula for $\mathcal{R}_i(\vec{h}_i)$, we need to provide a lower bound $\mathcal{R}_{i,\text{lb}}(\vec{h}_i)$ on $\mathcal{R}_i(\vec{h}_i)$. Subsequently, using $\mathcal{R}_{i,\text{lb}}(\vec{h}_i)$, we derive a lower bound on the outage capacity of

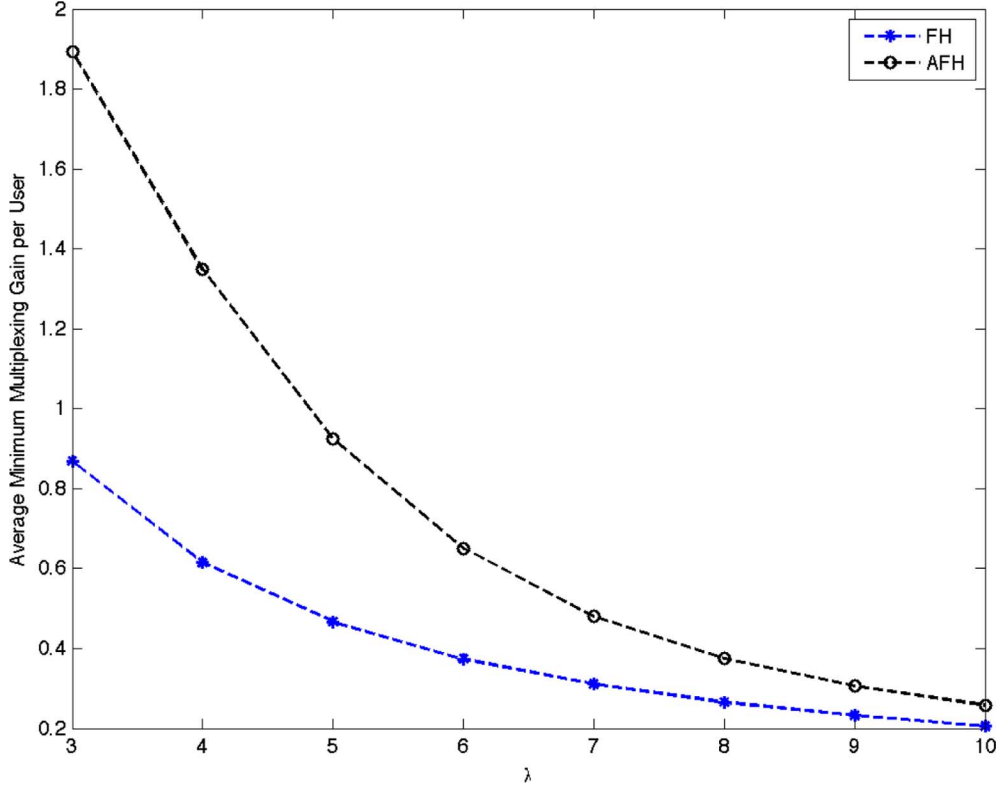


Fig. 3. Plots of $\eta_{\text{AFH}}^{(2)}$ and $\eta_{\text{FH}}^{(2)}$ in terms of λ for $u = 10$.

the i th user as $\sup\{R : \Pr\{\vec{h}_i : \mathcal{R}_{i,\text{lb}}(\vec{h}_i) < R\} < \varepsilon\}$, and show that this lower bound is higher than the actual outage capacity in the FD scheme in many scenarios.

We have already developed a lower bound on $\mathcal{R}_i(\vec{h}_i)$ in Theorem 2. In the following, we improve this lower bound to obtain better results in terms of outage capacity at a certain finite SNR level.

A. Lower Bounds on $\mathcal{R}_i(\vec{h}_i)$

Following the same lines as in the proof of Theorem 2

$$\mathcal{R}_i(\vec{h}_i) \geq v \log \left(\frac{\pi e |\mathbf{h}_{i,i}|^2 P}{v} 2^{-h(Z_{i,1})} + 1 \right). \quad (62)$$

We start with the following Lemma.

Lemma 2: There exists an upper bound on $h(Z_{i,1})$ given by

$$h(Z_{i,1}) \leq \sum_{l=0}^{\mathfrak{L}_i} \mathbf{a}_{i,l} \log(\pi e (\mathbf{c}_{i,l} P + \sigma^2)) + \mathfrak{H}_i - \mathfrak{G}_i \quad (63)$$

where

$$\mathfrak{H}_i \triangleq - \sum_{l=0}^{\mathfrak{L}_i} \mathbf{a}_{i,l} \log \mathbf{a}_{i,l} \quad (64)$$

and

$$\mathfrak{G}_i \triangleq \frac{1}{\mathbf{c}_{i,\mathfrak{L}_i} \gamma + 1} \sum_{l=1}^{\mathfrak{L}_i} \mathbf{a}_{i,l} \log \left(1 + \frac{(\mathbf{c}_{i,\mathfrak{L}_i} \gamma + 1) \sum_{m=0}^{l-1} \mathbf{a}_{i,m}}{\mathbf{a}_{i,l}} \right). \quad (65)$$

Proof: See Appendix A. ■

Applying Lemma 2 to (62), we obtain a lower bound on $\mathcal{R}_i(\vec{h}_i)$ as

$$\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i) \triangleq v \log \left(\frac{2^{-\mathfrak{H}_i} 2^{\mathfrak{G}_i} |\mathbf{h}_{i,i}|^2 \gamma}{v \prod_{l=1}^{\mathfrak{L}_i} (\mathbf{c}_{i,l} \gamma + 1)^{\mathbf{a}_{i,l}}} + 1 \right). \quad (66)$$

We make the following observations.

Observation 3: It can be immediately verified that $\mathfrak{H}_i$ does not depend on the crossover gains. However, $\mathfrak{G}_i$ is implicitly a function of all crossover gains as the partial sums $\sum_{m=1}^{l-1} \mathbf{a}_{i,m}$ for $2 \leq l \leq \mathfrak{L}_i$ depend on the ordering of the crossover gains. This will be investigated more in Lemma 3.

Observation 4: Since $\prod_{l=1}^{\mathfrak{L}_i} (\mathbf{c}_{i,l} \gamma + 1)^{\mathbf{a}_{i,l}} \leq \prod_{l=1}^{\mathfrak{L}_i} (\mathbf{c}_{i,\mathfrak{L}_i} \gamma + 1)^{\mathbf{a}_{i,l}} = (\mathbf{c}_{i,\mathfrak{L}_i} \gamma + 1)^{(1-\mathbf{a}_{i,0})}$, one obtains a looser version of $\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i)$ given by

$$\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i) \triangleq v \log \left(\frac{2^{-\mathfrak{H}_i} 2^{\mathfrak{G}_i} |\mathbf{h}_{i,i}|^2 \gamma}{v (\mathbf{c}_{i,\mathfrak{L}_i} \gamma + 1)^{1-\mathbf{a}_{i,0}}} + 1 \right). \quad (67)$$

We note that $\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i)$ still has the same asymptotic expression as that of $\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i)$ in the high SNR regime. As we will see later, the computational complexity of the lower bound on the outage capacity obtained by using $\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i)$ is much lower than that of $\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i)$.

As explained before, $\mathfrak{G}_i$ depends on the ordering of $\{\mathbf{c}_{i,l}\}_{l=0}^{\mathfrak{L}_i}$, which requires analyzing the order statistics of the channel gains. To avoid this, the following Lemma introduces a lower bound on $\mathfrak{G}_i$ that only depends on $\mathbf{c}_{i,\mathfrak{L}_i}$.

Lemma 3: Let $\mathfrak{B}$ be a Binomial random variable with parameters $(\mathfrak{N} - 1, p)$ where $p = \frac{v}{u}$. Then

$$\mathfrak{H}_i = (\mathfrak{N} - 1)\mathcal{H}(p) \quad (68)$$

and $\mathfrak{G}_i \geq \mathfrak{G}_{i,\text{lb}}$ where

$$\mathfrak{G}_{i,\text{lb}} \triangleq \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \left(E_{\mathfrak{B}} \{ \log(1 + (1 - p^{\mathfrak{B}}) \mathfrak{c}_{i,\mathfrak{L}_i}\gamma) \} - (\mathfrak{N} - 1)p \log p \right). \quad (69)$$

Proof: As $\mathfrak{L}_i = 2^{\mathfrak{N}-1} - 1$ with probability 1 and each user selects a certain frequency subband with probability p , the collection $\{\mathfrak{a}_{i,l}\}_{l=0}^{\mathfrak{L}_i}$ consists of the numbers $p^j(1-p)^{\mathfrak{N}-1-j}$ repeated $\binom{\mathfrak{N}-1}{j}$ times for $0 \leq j \leq \mathfrak{N} - 1$. Hence

$$\begin{aligned} \mathfrak{H}_i &= - \sum_{j=0}^{\mathfrak{N}-1} \Pr\{\mathfrak{B} = j\} \log(p^j(1-p)^{\mathfrak{N}-1-j}) \\ &= -(\mathfrak{N} - 1)p \log p - (\mathfrak{N} - 1 - (\mathfrak{N} - 1)p) \log(1 - p) \\ &= -(\mathfrak{N} - 1)(p \log p + (1 - p) \log(1 - p)). \end{aligned} \quad (70)$$

As for $\mathfrak{G}_i$, computation of $\sum_{m=0}^{l-1} \mathfrak{a}_{i,m}$ is not an easy task. In fact, it depends on the ordering of the crossover gains. For example, if $\mathfrak{N} = 4$, $l = 4$ and $i = 1$, this fact is shown in (71) at the bottom of the page.

To avoid this difficulty in describing $\mathfrak{G}_i$, we derive a lower bound on this quantity, which is not sensitive to the ordering of crossover gains. Taking each $\mathfrak{a}_{i,l}$, there exists a $0 \leq n \leq \mathfrak{N} - 1$ such that $\mathfrak{a}_{i,l} = p^n(1-p)^{\mathfrak{N}-1-n}$. This implies that $\mathfrak{a}_{i,l}$ corresponds to the interference plus noise power level $\frac{\sum_{k=1}^n |\mathfrak{h}_{j_k,i}|^2}{v} P + \sigma^2$ for some $1 \leq j_1 < \dots < j_n \leq \mathfrak{N}$ where $j_k \neq i$ for $1 \leq k \leq n$. Since $\frac{\sum_{k=1}^n |\mathfrak{h}_{j_k,i}|^2}{v} P + \sigma^2 > \frac{\sum_{t \in \mathcal{A} \subseteq \{1,2,\dots,n\}} |\mathfrak{h}_{j_t,i}|^2}{v} P + \sigma^2$ for any set $\mathcal{A} \subsetneq \{1,2,\dots,n\}$, and $\frac{\sum_{t \in \mathcal{A} \subseteq \{1,2,\dots,n\}} |\mathfrak{h}_{j_t,i}|^2}{v} P + \sigma^2$ is itself a power level in the pdf of the noise plus interference on each frequency subband, we conclude that its associated probability $p^{|\mathcal{A}|}(1-p)^{\mathfrak{N}-1-|\mathcal{A}|}$ is an element in the sequence $(\mathfrak{a}_{i,0}, \mathfrak{a}_{i,1}, \dots, \mathfrak{a}_{i,l-1})$. Therefore, we come up with the following lower bound:

$$\begin{aligned} \sum_{m=0}^{l-1} \mathfrak{a}_{i,m} &\geq \sum_{\mathcal{A} \subseteq \{1,2,\dots,n\}} p^{|\mathcal{A}|}(1-p)^{\mathfrak{N}-1-|\mathcal{A}|} \\ &= \sum_{n'=0}^{n-1} \binom{n}{n'} p^{n'} (1-p)^{\mathfrak{N}-1-n'}. \end{aligned} \quad (72)$$

Using (72) in (65), one can develop a lower bound on $\mathfrak{G}_i$ as in (73) at the bottom of the page. This concludes the Lemma. ■

$$\sum_{m=0}^3 \mathfrak{a}_{1,m} = \begin{cases} (1-p)^3 + 2p(1-p)^2 + p^2(1-p) & \text{if } |\mathfrak{h}_{2,1}|^2 < |\mathfrak{h}_{3,1}|^2 < |\mathfrak{h}_{2,1}|^2 + |\mathfrak{h}_{3,1}|^2 < |\mathfrak{h}_{4,1}|^2 \\ (1-p)^3 + 3p(1-p)^2 & \text{if } |\mathfrak{h}_{2,1}|^2 < |\mathfrak{h}_{3,1}|^2 < |\mathfrak{h}_{4,1}|^2 < |\mathfrak{h}_{2,1}|^2 + |\mathfrak{h}_{3,1}|^2 \end{cases}. \quad (71)$$

$$\begin{aligned} \mathfrak{G}_i &\geq \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \sum_{n=1}^{\mathfrak{N}-1} \binom{\mathfrak{N}-1}{n} p^n (1-p)^{\mathfrak{N}-1-n} \log \left(1 + \frac{(\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1) \sum_{n'=0}^{n-1} \binom{n}{n'} p^{n'} (1-p)^{\mathfrak{N}-1-n'}}{p^n (1-p)^{\mathfrak{N}-1-n}} \right) \\ &= \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \sum_{n=1}^{\mathfrak{N}-1} \binom{\mathfrak{N}-1}{n} p^n (1-p)^{\mathfrak{N}-1-n} \log \left(1 + \frac{(\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1) \sum_{n'=0}^{n-1} \binom{n}{n'} p^{n'} (1-p)^{n-n'}}{p^n} \right) \\ &= \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \sum_{n=1}^{\mathfrak{N}-1} \binom{\mathfrak{N}-1}{n} p^n (1-p)^{\mathfrak{N}-1-n} \log \left(1 + \frac{(1-p^n)(\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1)}{p^n} \right) \\ &= - \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \sum_{n=0}^{\mathfrak{N}-1} \binom{\mathfrak{N}-1}{n} n p^n (1-p)^{\mathfrak{N}-1-n} \log p \\ &\quad + \frac{1}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1} \sum_{n=0}^{\mathfrak{N}-1} \binom{\mathfrak{N}-1}{n} p^n (1-p)^{\mathfrak{N}-1-n} \log(1 + (1-p^n) \mathfrak{c}_{i,\mathfrak{L}_i}\gamma) \\ &= \frac{E_{\mathfrak{B}} \{ \log(1 + (1-p^{\mathfrak{B}}) \mathfrak{c}_{i,\mathfrak{L}_i}\gamma) \} - (\mathfrak{N} - 1)p \log p}{\mathfrak{c}_{i,\mathfrak{L}_i}\gamma + 1}. \end{aligned} \quad (73)$$

From now on, we replace $\mathcal{G}_i$ by $\mathcal{G}_{i,\text{lb}}$ defined in (69) in all expressions offered for the lower bounds on $\mathcal{R}_i(\vec{h}_i)$. We denote $\alpha_{i,0}$, $\mathfrak{H}_i$ and $\mathcal{G}_{i,\text{lb}}$ by $a(v, \mathfrak{N}) = (1 - \frac{v}{u})^{\mathfrak{N}-1}$, $\mathfrak{H}(v, \mathfrak{N})$ and $\mathcal{G}_{i,\text{lb}}(v, \mathfrak{N})$, respectively,¹¹ to emphasize their dependence on v , $\mathfrak{N}$.

As a special case, let us assume $v = u$, i.e., all users spread their power on the whole spectrum. This scheme is called Full-Band Spreading (FBS). In this case, it can be observed that $\alpha_{i,l} = 0$ for $l \leq \mathfrak{L}_i - 1$ and $\alpha_{i,\mathfrak{L}_i} = 1$. This yields $\mathfrak{a}(u, \mathfrak{N}) = \mathfrak{H}(u, \mathfrak{N}) = \mathcal{G}_{i,\text{lb}}(u, \mathfrak{N}) = 0$. In fact, $\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i)$ is tight for $v = u$, i.e., $\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i)$ is exactly the achievable rate of the i th user when all users transmit over the whole spectrum. We denote this rate by $\mathcal{R}_{i,\text{FBS}}(\vec{h}_i)$, which is given by

$$\mathcal{R}_{i,\text{FBS}}(\vec{h}_i) = u \log \left(\frac{|\mathfrak{h}_{i,i}|^2 \gamma}{u \left(\frac{\mathfrak{c}_{i,\mathfrak{L}_i} \gamma}{u} + 1 \right)} + 1 \right). \quad (74)$$

Observation 5: A straightforward method to develop a lower bound on the achievable rate in an additive white non-Gaussian noise channel is to replace the noise with a Gaussian noise of the same covariance matrix [61]. Following this approach, it is easy to derive the following lower bound on $\mathcal{R}_i(\vec{h}_i)$:

$$\mathcal{R}_{i,g}(\vec{h}_i) \triangleq v \log \left(\frac{|\mathfrak{h}_{i,i}|^2 \gamma}{v \left(\frac{\mathfrak{c}_{i,\mathfrak{L}_i} \gamma}{u} + 1 \right)} + 1 \right) \quad (75)$$

where the index “ g ” stands for Gaussian. There are two facts that are worth mentioning about $\mathcal{R}_{i,g}(\vec{h}_i)$. First, it is seen that $\lim_{\gamma \rightarrow \infty} \mathcal{R}_{i,g}(\vec{h}_i) < \infty$. Another point is that $\mathcal{R}_{i,g}(\vec{h}_i)$ is an increasing function of v . However, setting $v = u$ in the expression of $\mathcal{R}_{i,g}(\vec{h}_i)$ yields the expression of $\mathcal{R}_{i,\text{FBS}}(\vec{h}_i)$. Therefore, for all realizations of the channel gains and all ranges of γ

$$\mathcal{R}_{i,g}(\vec{h}_i) \leq \mathcal{R}_{i,\text{FBS}}(\vec{h}_i). \quad (76)$$

This indicates that using $\mathcal{R}_{i,g}(\vec{h}_i)$ as a lower bound on the achievable rate of users in the FH scheme provides no proof of advantage for FH over FBS.

B. Lower Bounds on $R_{\text{FH}}(\varepsilon)$ and System Design

Based on the preceding discussion, we can derive lower bounds on $R_{\text{FH}}(\varepsilon)$, namely $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$ associated with the lower bounds $\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i)$ and $\mathcal{R}_{i,\text{lb}}^{(2)}(\vec{h}_i)$, respectively. Consequently, we can obtain estimates of the optimum hopping

¹¹We note that $\alpha_{i,0}$ and $\mathfrak{H}_i$ do not depend on i , however, $\mathcal{G}_{i,\text{lb}}$ depends on i through $\sum_{j \neq i} |\mathfrak{h}_{j,i}|^2$.

parameter¹² by maximizing $R_{\text{FH}}^{(1)}(\varepsilon)$ or $R_{\text{FH}}^{(2)}(\varepsilon)$ over v . In Sections VII-B–VII-E, we separately compute $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$.

1) *Computation of $R_{\text{FH}}^{(1)}(\varepsilon)$:* We start with the following definitions.

Definition 1: Let $n \in \mathbb{N}$. For $p \in [0, 1]$ and $b > 0$, the function $\alpha_n(\cdot; b, p) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is defined by

$$\alpha_n(\theta; b, p) \triangleq \frac{E_{\mathfrak{B}} \{ \log(1 + (1 - p^{\mathfrak{B}})b\theta) \} - (n-1)p \log p}{b\theta + 1} \quad (77)$$

where $\mathfrak{B}$ is a Binomial random variable with parameters $(n-1, p)$. $\square$

Definition 2: Let $n \geq 2$ be an integer. For $b_1 < 0$, $b_2 > 0$ and $p \in [0, 1]$, we define $\psi_n(b_1, b_2, p)$ as in (78) at the bottom of the page, where for each m , $\{\theta_{m,m'}\}_{m'=1}^{n-1}$ consists of all possible summations of m elements in the set of dummies $\{\theta_i\}_{i=1}^{n-1}$ and $\beta_{m,n}(p) \triangleq p^m(1-p)^{n-1-m}$. $\square$

The following Proposition offers an expression to compute $R_{\text{FH}}^{(1)}(\varepsilon)$.

Proposition 2:

$$R_{\text{FH}}^{(1)}(\varepsilon) = \sup \{ R : \Xi_1(R) > 1 - \varepsilon \} \quad (79)$$

where

$$\Xi_1(R) = q_1 e^{\frac{v}{\gamma} \left(1 - 2^{\frac{R}{v}} \right)} + \sum_{n=2}^{n_{\max}} q_n \psi_n(b_{1,n}, b_{2,n}, p) \quad (80)$$

$$b_{1,n} = \frac{2^{\mathfrak{H}(v,n)} \left(1 - 2^{\frac{R}{v}} \right) v}{\gamma}, \quad b_{2,n} = \frac{\gamma}{v} \quad \text{and} \quad p = \frac{v}{u}.$$

Proof: See Appendix D. $\blacksquare$

The expression given in (79) is quite complicated. On one hand, the multiple integrals do not have a closed form. On the other hand, the maximization $\max_v R_{\text{FH}}^{(1)}(\varepsilon)$ must be computed numerically. However, $R_{\text{FH}}^{(1)}(\varepsilon)$ is the best lower bound on $R_{\text{FH}}(\varepsilon)$ as $\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{h}_i)$ is the best lower bound we have found on the achievable rate of the i th user in the FH scenario.

2) *Computation of $R_{\text{FH}}^{(2)}(\varepsilon)$:* We start with the following definition.

Definition 3: Let $n \geq 2$ be an integer. For $b_1 < 0$, $b_2 > 0$ and $p_1, p_2 \in [0, 1]$, we define $\phi_n(b_1, b_2, p_1, p_2)$ as in (81) at the bottom of the page.

¹²The optimum hopping parameter is the value of v that maximizes $R_{\text{FH}}(\varepsilon)$.

$$\psi_n(b_1, b_2, p) \triangleq \int_{\theta_1 \geq 0, \dots, \theta_{n-1} \geq 0} \exp \left(b_1 2^{-\alpha_n(\theta_{n-1,1}; b_2, p)} \prod_{m=1}^{n-1} \prod_{m'=1}^{n-1} (b_2 \theta_{m,m'} + 1)^{\beta_{m,n}(p)} - \theta_{n-1,1} \right) d\theta_1 \cdots d\theta_{n-1} \quad (78)$$

$$\phi_n(b_1, b_2, p_1, p_2) \triangleq \frac{1}{(n-2)!} \int_0^\infty \theta^{n-2} \exp \left(b_1 (b_2 \theta + 1)^{p_1} 2^{-\alpha_n(\theta; b_2, p_2)} - \theta \right) d\theta. \quad (81)$$

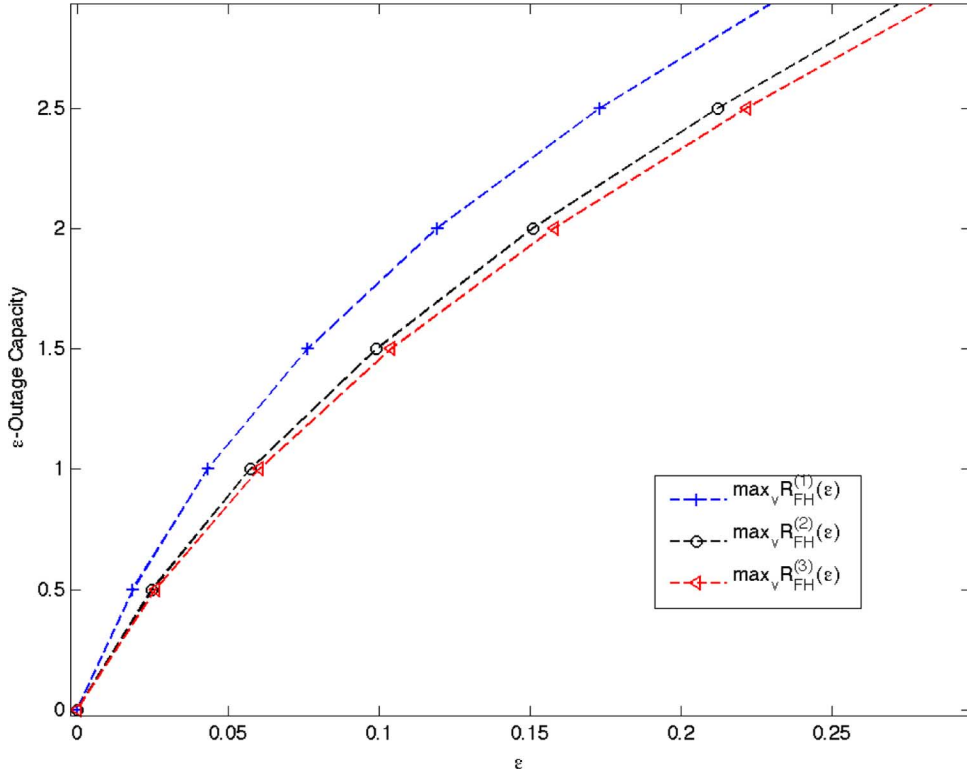


Fig. 4. Sketch of $\max_v R_{\text{FH}}^{(k)}(\varepsilon)$ (bits/transmission slot) for $k = 1, 2, 3$ in a setup where $(q_1, q_2, q_3, q_4) = (0.4, 0.2, 0.2, 0.2)$, $u = 8$, and $\gamma = 20$ dB.

Using this class of functions, the following Proposition yields $R_{\text{FH}}^{(2)}(\varepsilon)$.

Proposition 3:

$$R_{\text{FH}}^{(2)}(\varepsilon) = \sup \{R : \Xi_2(R) > 1 - \varepsilon\} \quad (82)$$

where

$$\Xi_2(R) = q_1 e^{\frac{v}{\gamma} \left(1 - 2^{\frac{R}{v}}\right)} + \sum_{n=2}^{n_{\max}} q_n \phi_n(b_{1,n}, b_2, p_{1,n}, p_2) \quad (83)$$

$p_{1,n} = 1 - a(v, n)$, $p_2 = \frac{v}{u}$ and $b_{1,n}$ and b_2 are given in Proposition 2.

Proof: The proof follows from similar steps along the lines of the proof for Proposition 2. ■

Comparing the expressions for $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$, it can be observed that computation of $R_{\text{FH}}^{(2)}(\varepsilon)$ involves only single integrations, while the computation of $R_{\text{FH}}^{(1)}(\varepsilon)$ involves iterated integrations that are not tractable for many cases. To further reduce the complexity of computation, the following Corollary yields another lower bound on $R_{\text{FH}}(\varepsilon)$ that involves no numerical integrations. We denote this lower bound by $R_{\text{FH}}^{(3)}(\varepsilon)$.

Corollary 1: Let

$$R_{\text{FH}}^{(3)}(\varepsilon) \triangleq \sup \{R : \Xi_3(R) > 1 - \varepsilon\} \quad (84)$$

where

$$\Xi_3(R) = \sum_{n=1}^{n_{\max}} q_n \exp \left(b_{1,n} ((n-1)b_2 + 1)^{1-a(v,n)} \right) \quad (85)$$

and $b_{1,n}$ and b_2 are given in Proposition 2. Then

$$R_{\text{FH}}(\varepsilon) \geq R_{\text{FH}}^{(1)}(\varepsilon) \geq R_{\text{FH}}^{(2)}(\varepsilon) \geq R_{\text{FH}}^{(3)}(\varepsilon). \quad (86)$$

Proof: See Appendix E. ■

Fig. 4 shows the three lower bounds $\max_v R_{\text{FH}}^{(k)}(\varepsilon)$ on $\max_v R_{\text{FH}}(\varepsilon)$ for $k = 1, 2, 3$ in a system where, at most, four users become active simultaneously with $(q_1, q_2, q_3, q_4) = (0.4, 0.2, 0.2, 0.2)$, $u = 8$ and $\gamma = 20$ dB. As can be observed from this figure, $\max_v R_{\text{FH}}^{(2)}(\varepsilon)$ and $\max_v R_{\text{FH}}^{(3)}(\varepsilon)$ are pretty close to each other while they are within a considerable gap to $\max_v R_{\text{FH}}^{(1)}(\varepsilon)$, especially for larger values of ε . It is notable that the maximization over v is performed separately for each ε .

C. Asymptotic Analysis

Having the expression for the ε -outage capacity, we can find the best operational point of the system in terms of v , the number of selected subbands. For this purpose, we consider some asymptotic cases in terms of ε and γ and discuss the optimum value of v in these regimes. As the expressions of $R_{\text{FH}}^{(1)}(\varepsilon)$ and $R_{\text{FH}}^{(2)}(\varepsilon)$ are not analytically tractable, we use $R_{\text{FH}}^{(3)}(\varepsilon)$ for our analysis. For simulation purposes, we use $R_{\text{FH}}^{(1)}(\varepsilon)$ or $R_{\text{FH}}^{(2)}(\varepsilon)$.

1) *Asymptotically Small ε :* In this case, one can easily show that $\lim_{\varepsilon \rightarrow 0} R_{\text{FH}}^{(3)}(\varepsilon) = 0$. Therefore, in (84), we can approximate $\frac{(1-2^{\frac{R}{v}})^v}{\gamma}$ by $-\frac{R \ln 2}{\gamma}$ and hence

$$e^{\frac{v}{\gamma} \left(1 - 2^{\frac{R}{v}}\right)} \approx 1 - \frac{R \ln(2)}{\gamma}. \quad (87)$$

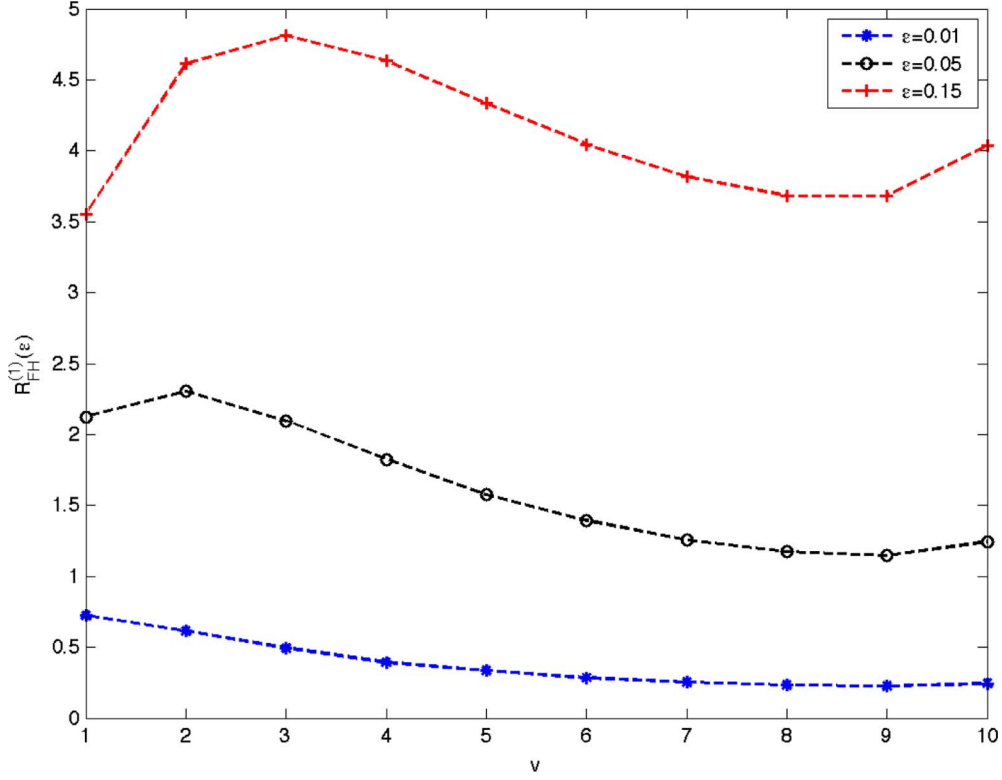


Fig. 5. Sketch of $R_{\text{FH}}^{(1)}(\varepsilon)$ (bits/transmission slot) for different values of ε in a network with $(q_1, q_2) = (0.5, 0.5)$, $u = 10$, and $\gamma = 20$ dB.

By the same token, the term on the right-hand side (RHS) of (84) can be approximated as

$$\begin{aligned} & \sum_{n=1}^{n_{\max}} q_n \exp\left(b_{1,n}((n-1)b_2 + 1)^{1-a(v,n)}\right) \\ & \approx \sum_{n=1}^{n_{\max}} q_n \exp\left(-\frac{R \ln 2}{\gamma} 2^{\mathfrak{H}(v,n)} \left(\frac{n-1}{v}\gamma + 1\right)^{1-a(v,n)}\right) \\ & \stackrel{(a)}{\approx} \sum_{n=1}^{n_{\max}} q_n \left(1 - \frac{R \ln 2}{\gamma} f(v, n, \gamma)\right) \end{aligned} \quad (88)$$

where (a) follows from the fact that the term

$$f(v, n, \gamma) \triangleq 2^{\mathfrak{H}(v,n)} \left(\frac{n-1}{v}\gamma + 1\right)^{1-a(v,n)} \quad (89)$$

does not depend on ε . Using (88) in (84) yields

$$R_{\text{FH}}^{(3)}(\varepsilon) \approx \frac{\varepsilon \gamma}{\left(\sum_{n=1}^{n_{\max}} q_n f(v, n, \gamma)\right) \ln 2}. \quad (90)$$

One can observe that the function $f(v, n, \gamma)$ is concave in terms of v for all $n \geq 1$. Hence, the function $g(v) \triangleq \sum_{n=1}^{n_{\max}} q_n f(v, n, \gamma)$ is concave as well. As such, the minimum of $g(v)$ occurs either at $v = 1$ or $v = u$. In particular, for $v = 1$ to be the optimum value, we must have $g(1) < g(u)$ or equivalently

$$\sum_{n=1}^{n_{\max}} \frac{q_n u^{n-1} ((n-1)\gamma + 1)^{1-(1-\frac{1}{u})^{n-1}}}{(u-1)(1-\frac{1}{u})^{(n-1)}}$$

$$\begin{aligned} & < \sum_{n=1}^{n_{\max}} q_n \left(1 + \frac{(n-1)\gamma}{u}\right) \\ & = 1 + \frac{E\{N\} - 1}{u} \gamma \end{aligned} \quad (91)$$

where we have used the fact that $2^{\mathfrak{H}(1,n)} = \left(\frac{u}{(u-1)^{1-\frac{1}{u}}}\right)^{n-1}$.

Fig. 5 offers the curves of $R_{\text{FH}}^{(1)}(\varepsilon)$ in terms of v for $\varepsilon = 0.01$, 0.05, and 0.15. The underlying network is characterized with $q_1 = q_2 = 0.5$, $u = 10$, and $\gamma = 20$ dB. It is seen that for $\varepsilon = 0.01$, taking $v = 1$ yields the best performance. As far as $q_1 = q_2 = 0.5$ and for sufficiently small ε , the (91) simplifies to $\frac{u^u}{(u-1)^{u-1}} < \frac{1}{1+\gamma} \left(1 + \frac{\gamma}{u}\right)^u$. Setting $u = 10$ yields 7.052 dB as the minimum SNR value that guarantees $v = 1$ is the best choice. Since $\gamma = 20$ dB $>$ 7.052 dB, we expect $v = 1$ is the best choice, which is confirmed in the plot. However, as ε increases, we move away from the *asymptotically small* ε region and $v = 1$ is no longer an optimal choice.

2) *Asymptotically Small γ* : In this case, one can easily show that $\lim_{\gamma \rightarrow 0} R_{\text{FH}}^{(3)}(\varepsilon) = 0$. Therefore, similar to the previous case, we can use the approximation

$$\frac{(1 - 2^{\frac{R}{\gamma}})v}{\gamma} \approx -\frac{R \ln 2}{\gamma}. \quad (92)$$

Defining $\tau \triangleq \exp\left(-\frac{R \ln 2}{\gamma}\right)$, one can rewrite (84) as

$$R_{\text{FH}}^{(3)}(\varepsilon) \approx \sup \left\{ R : \sum_{n=1}^{n_{\max}} q_n \tau^{f(v,n,\gamma)} > 1 - \varepsilon \right\} \quad (93)$$

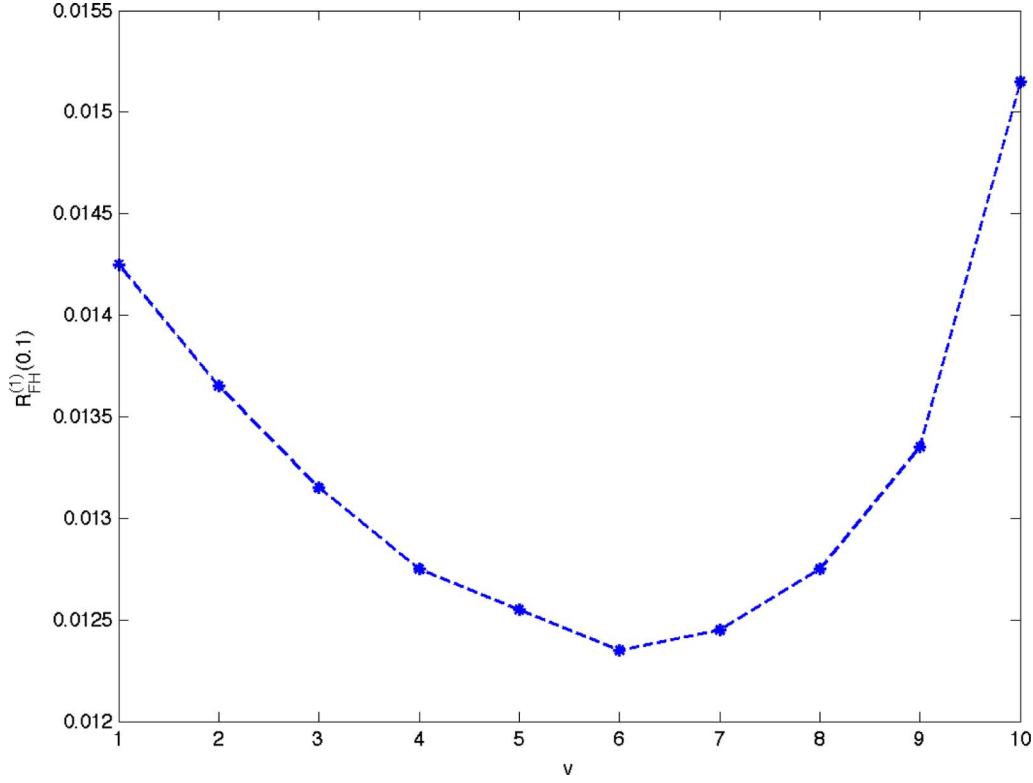


Fig. 6. Sketch of $R_{\text{FH}}^{(1)}(0.1)$ (bits/transmission slot) in a network with $(q_1, q_2) = (0.5, 0.5)$, $\gamma = -10$ dB, and $u = 10$.

where $f(v, n, \gamma)$ is given in (89). As τ is a decreasing function in terms of R , we can write

$$R_{\text{FH}}^{(3)}(\varepsilon) = -\frac{\gamma \ln(\tau^*)}{\ln 2} \quad (94)$$

where

$$\tau^* = \inf \left\{ \tau : \sum_{n=1}^{n_{\max}} q_n \tau^{f(v, n, \gamma)} > 1 - \varepsilon \right\}. \quad (95)$$

Therefore

$$\max_v R_{\text{FH}}^{(3)}(\varepsilon) \approx -\frac{\gamma}{\ln 2} \ln \left(\min_v \tau^* \right). \quad (96)$$

It can be shown that $\min_v \tau^*$ occurs when $f(v, n, \gamma)$ takes its minimum value over v for each n . But, as $\gamma \ll 1$, the term $\left(\frac{n-1}{v}\gamma + 1\right)^{1-a(v, n)} \approx 1$ and, hence, $f(v, n, \gamma) \approx 2^{\mathfrak{H}(v, n)}$, which is uniformly minimized for all values of n by taking $v = u$. This gives $\tau^* = 1 - \varepsilon$, and, hence

$$R_{\text{FH}}^{(3)}(\varepsilon) \approx -\gamma \log(1 - \varepsilon). \quad (97)$$

This is exactly the outage capacity of a point-to-point system without interference. Therefore, in the low SNR regime, interference has no destructive effect on the outage capacity justifying the optimality of $v = u$.

Fig. 6 presents the plot of $R_{\text{FH}}^{(1)}(0.1)$ versus v for $\gamma = -10$ dB in a system with $q_1 = q_2 = 0.5$ and $u = 10$. It is seen that $v = u = 10$ is the best choice, which is expected by our analysis. Also, we observe that the outage capacity is not very sensitive to the value of v .

3) *Asymptotically High γ* : Recalling the expression of $R_{\text{FH}}^{(3)}(\varepsilon)$ given in (84), we have

$$R_{\text{FH}}^{(3)}(\varepsilon) = \sup \left\{ R : \sum_{n=1}^{n_{\max}} q_n e^{-\frac{v f(v, n, \gamma)(2^{R/v} - 1)}{\gamma}} > 1 - \varepsilon \right\}. \quad (98)$$

As $\gamma \rightarrow \infty$, the term $f(v, n, \gamma)$ grows polynomially with γ for any $n \geq 1$. The outage event is determined by the term with the maximum $f(v, n, \gamma)$. In fact, since $f(v, n, \gamma)$ is an increasing function in terms of n , it follows that $\sum_{n=1}^{n_{\max}} q_n e^{-\frac{v f(v, n, \gamma)(2^{R/v} - 1)}{\gamma}} \approx \sum_{n=1}^{n_{\max}-1} q_n + q_{n_{\max}} e^{-\frac{v f(v, n_{\max}, \gamma)(2^{R/v} - 1)}{\gamma}}$. Therefore, (98) simplifies to

$$R_{\text{FH}}^{(3)}(\varepsilon) \approx \sup \left\{ R : e^{-\frac{v f(v, n_{\max}, \gamma)(2^{R/v} - 1)}{\gamma}} > 1 - \frac{\varepsilon}{q_{n_{\max}}} \right\} \quad (99)$$

which gives

$$\begin{aligned} R_{\text{FH}}^{(3)}(\varepsilon) &\approx v \log \left(1 - \frac{\gamma \ln \left(1 - \frac{\varepsilon}{q_{n_{\max}}} \right)}{v f(v, n_{\max}, \gamma)} \right) \\ &= v \log \left(1 - \frac{\gamma 2^{-\mathfrak{H}(v, n_{\max})} \ln \left(1 - \frac{\varepsilon}{q_{n_{\max}}} \right)}{v \left(1 + \frac{n_{\max}-1}{v} \gamma \right)^{1-(1-\frac{v}{u})^{n_{\max}-1}}} \right) \\ &\stackrel{(a)}{\approx} v \log \left(1 - \kappa_v \gamma^{(1-\frac{v}{u})^{n_{\max}-1}} \ln \left(1 - \frac{\varepsilon}{q_{n_{\max}}} \right) \right) \end{aligned} \quad (100)$$

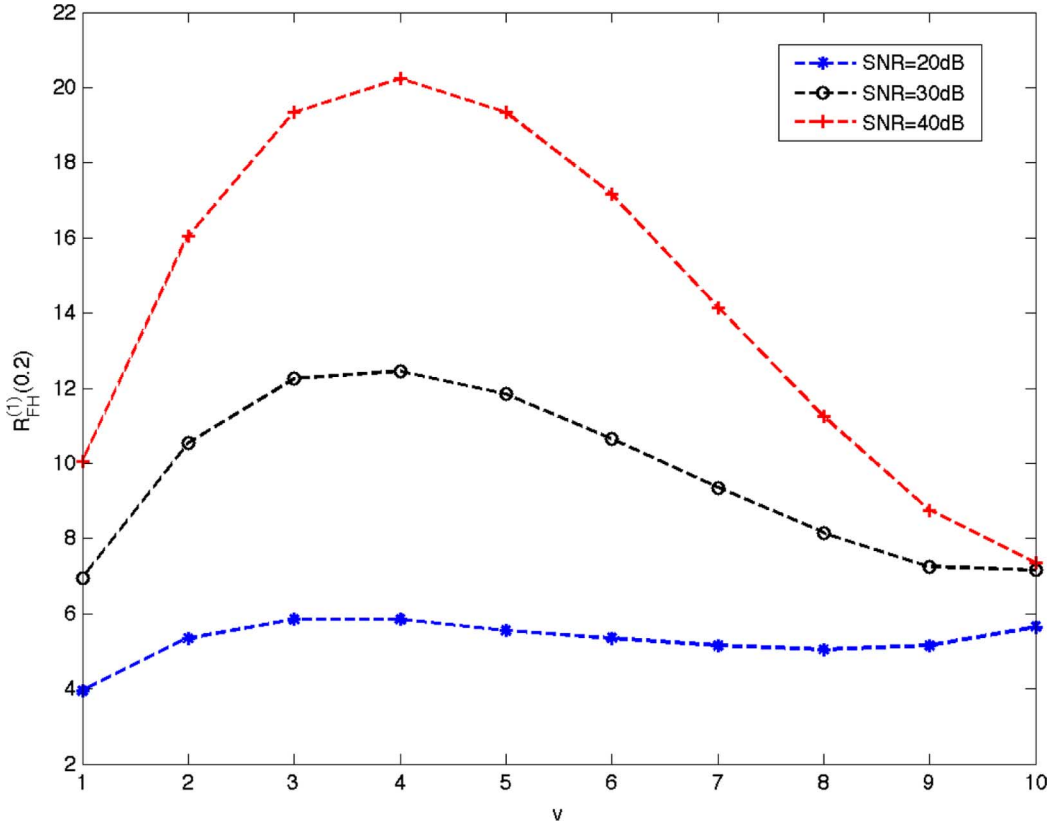


Fig. 7. Sketch of $R_{\text{FH}}^{(1)}(0.2)$ (bits/transmission slot) at different SNR levels in a network with $n_{\max} = 2$, $(q_1, q_2) = (0.5, 0.5)$, and $u = 10$.

where

$$\kappa_v \triangleq \frac{2^{-\mathcal{H}(v, n_{\max})}}{v} \left(\frac{n_{\max} - 1}{v} \right)^{\left(1 - \frac{v}{u}\right)^{n_{\max} - 1} - 1} \quad (101)$$

and (a) follows from the assumption that γ lies in the high SNR range. It is observed that the maximization of the above expression with respect to v is equivalent to the maximization of the term $v \left(1 - \frac{v}{u}\right)^{n_{\max} - 1}$ with respect to v , as SNR tends to infinity. This yields

$$v = \left\lceil \frac{u}{n_{\max}} \right\rceil. \quad (102)$$

Fig. 7 shows the curves of $R_{\text{FH}}^{(1)}(0.2)$ versus v for different values of SNR in a system with parameters $n_{\max} = 2$, $(q_1, q_2) = (0.5, 0.5)$ and $u = 10$. In general, for a sufficiently large, however finite, value of SNR, one can obtain the optimum value for v from

$$v = \arg \max_{1 \leq v \leq u} \left\{ v \log \left(1 - \kappa_v \gamma^{\left(1 - \frac{v}{u}\right)^{n_{\max} - 1}} \ln \left(1 - \frac{\varepsilon}{q_{n_{\max}}} \right) \right) \right\}. \quad (103)$$

For example, it is easy to verify that in a system with the above parameters at $\gamma = 40$ dB, one gets $v = 4$, while $\left\lceil \frac{u}{n_{\max}} \right\rceil = \left\lceil \frac{10}{2} \right\rceil = 5$. This is in agreement with the plot of $R_{\text{FH}}^{(1)}(0.2)$ given in Fig. 7 for $\gamma = 40$ dB.

D. Comparison With Other Schemes

In this section, we compare the performance of the proposed FH scenario with that of the FD scheme in terms of the ε -outage capacity. In the FD scheme, the spectrum is primarily divided into $n_{\text{des}} = \min\{n_{\max}, u\}$ bands and each active user only occupies one band. If $n_{\max} \leq u$ and all the $n_{\max}$ users are active all the time, i.e., $q_{n_{\max}} = 1$, this scheme results in the most efficient usage of the bandwidth. This makes the FD scenario superior to other schemes proposed in the literature, especially in the high SNR regime. However, in a practical situation, the number of concurrently active users is smaller than n_{des} with a nonzero probability. This makes FD highly inefficient on the heels that a considerable portion of the subbands is unused. In addition to FD, we also study the ε -outage capacity of the FBS scenario, which is a special case of FH. In fact, FD and FBS can be considered as two extreme spectrum management schemes where the former avoids any interference among the users, while the latter makes all users share the same spectrum all the time. In the sequel, we compute $R_{\text{FD}}(\varepsilon)$ and $R_{\text{FBS}}(\varepsilon)$.

1) *Computation of $R_{\text{FD}}(\varepsilon)$* : In the FD scenario, the spectrum is already divided into n_{des} nonoverlapping bands each containing $\frac{u}{n_{\text{des}}}$ subbands. Each user that becomes active occupies one of the units. As there is no interference among users, the outage event for the i th user can be written as

$$\mathcal{O}_{i,\text{FD}}(R) = \left\{ \mathbf{h}_{i,i} : \frac{u}{n_{\text{des}}} \log \left(1 + \frac{n_{\text{des}} |\mathbf{h}_{i,i}|^2 \gamma}{u} \right) < R \right\}. \quad (104)$$

As $h_{i,i}$ is a complex Gaussian random variable with variance $\frac{1}{2}$ per dimension, $|h_{i,i}|^2$ is an exponential random variable with parameter 1. Thus

$$\Pr\{\mathcal{O}_{i,\text{FD}}(R)\} = 1 - \exp\left(\frac{u\left(1 - 2^{\frac{n_{\text{des}}R}{u}}\right)}{n_{\text{des}}}\frac{1}{\gamma}\right) \quad (105)$$

and

$$\begin{aligned} R_{\text{FD}}(\varepsilon) &= \sup\{R : \Pr\{\mathcal{O}_{i,\text{FD}}(R)\} < \varepsilon\} \\ &= \sup\left\{R : \exp\left(\frac{u\left(1 - 2^{\frac{n_{\text{des}}R}{u}}\right)}{n_{\text{des}}}\frac{1}{\gamma}\right) > 1 - \varepsilon\right\} \\ &= \frac{u}{n_{\text{des}}} \log\left(1 - \frac{n_{\text{des}}\gamma \ln(1 - \varepsilon)}{u}\right). \end{aligned} \quad (106)$$

2) *Computation of $R_{\text{FBS}}(\varepsilon)$* : Using (74), the following Proposition yields $R_{\text{FBS}}(\varepsilon)$.

Proposition 4:

$$R_{\text{FBS}}(\varepsilon) = \sup\left\{R : e^{\frac{u}{\gamma}\left(1 - 2^{\frac{R}{u}}\right)} \sum_{n=1}^{n_{\text{max}}} q_n 2^{-\frac{(n-1)R}{u}} > 1 - \varepsilon\right\}. \quad (107)$$

Proof: The proof follows from similar steps along the lines of the proof for Proposition 2. ■

3) *Comparison of $R_{\text{FH}}(\varepsilon)$, $R_{\text{FD}}(\varepsilon)$, and $R_{\text{FBS}}(\varepsilon)$ in the Asymptotically Small ε Regime*: Noting that $\ln(1 - \varepsilon) \approx -\varepsilon$, we get $\log\left(1 - \frac{n_{\text{des}}\gamma \ln(1 - \varepsilon)}{u}\right) \approx \frac{n_{\text{des}}\gamma \varepsilon}{u \ln 2}$. Therefore

$$R_{\text{FD}}(\varepsilon) \approx \frac{\gamma \varepsilon}{\ln 2} \quad (108)$$

which is the maximum achievable ε -outage capacity in the underlying network for small values of ε . As for FBS, using (107) and noting that $\frac{R}{u} \ll 1$, we get

$$R_{\text{FBS}}(\varepsilon) \approx \frac{\varepsilon \gamma}{\left(1 + \frac{E\{N\}-1}{u}\right) \ln 2}. \quad (109)$$

Comparing (90) and (109) reveals that $R_{\text{FH}}^{(3)}(\varepsilon)$ is larger than $R_{\text{FBS}}(\varepsilon)$ in the low ε regime as far as (91) is satisfied. Furthermore, in the case that $E\{N\} \ll \frac{u}{\gamma}$, the above equation implies that the FBS scheme, and consequently the FH scenario, achieves the optimal performance of the FD scheme.

4) *Comparison of $R_{\text{FH}}(\varepsilon)$, $R_{\text{FD}}(\varepsilon)$, and $R_{\text{FBS}}(\varepsilon)$ in the Asymptotically Small γ Regime*: Using (106) and (107), it can be realized that for the asymptotically small γ

$$R_{\text{FD}}(\varepsilon) = R_{\text{FBS}}(\varepsilon) = -\gamma \log(1 - \varepsilon) \quad (110)$$

which is the same value obtained in the previous section for $R_{\text{FH}}(\varepsilon)$ (by setting $v = u$) and is the maximum achievable outage capacity in the network. Therefore, in this regime, spectrum division and spectrum sharing both achieve the optimal performance. Furthermore, the Gaussian lower bound given in (75) implies that the FH scheme is optimal in the low SNR

regime, regardless of v . However, if we use the proposed lower bounds (e.g., $R_{\text{FH}}^{(3)}(\varepsilon)$), due to the presence of the term $2^{\mathfrak{H}(v,n)}$ in the expression of such lower bounds, taking $v < u$ does not yield this conclusion.

5) *Comparison of $R_{\text{FH}}(\varepsilon)$, $R_{\text{FD}}(\varepsilon)$, and $R_{\text{FBS}}(\varepsilon)$ in the Asymptotically High γ Regime*: From (106)

$$\begin{aligned} R_{\text{FD}}(\varepsilon) &\approx \frac{u}{n_{\text{des}}} \log \gamma + \frac{u}{n_{\text{des}}} \log\left(-\frac{n_{\text{des}}}{u} \ln(1 - \varepsilon)\right) \\ &\approx \frac{u}{n_{\text{des}}} \log \gamma \end{aligned} \quad (111)$$

as $\gamma \rightarrow \infty$. Also, (107) indicates that $R_{\text{FBS}}(\varepsilon)$ saturates as γ tends to infinity and as such, FBS is highly inefficient in this regime. In fact

$$\lim_{\gamma \rightarrow \infty} R_{\text{FBS}}(\varepsilon) = \sup\left\{R : \sum_{n=1}^{n_{\text{max}}} q_n 2^{-\frac{(n-1)R}{u}} > 1 - \varepsilon\right\}. \quad (112)$$

To show the supremacy of FH over FD, we consider a practical scenario with $u = n_{\text{max}}$ subbands. Each user can be active with some probability p independently of other users. It is assumed that $n_{\text{max}} \gg 1$ and $p \ll 1$ such that $\lambda \triangleq p n_{\text{max}}$ is a constant. Let us assume that the i th user is active. The number of users, other than the i th user, which are simultaneously active together with the i th user is a Binomial random variable with parameters $(n_{\text{max}} - 1, p)$. This random variable, denoted by $\mathfrak{N}$, can be well approximated¹³ by a Poisson random variable with parameter $(n_{\text{max}} - 1)p \approx \lambda$.

For some $\hat{n} \geq 1$ where u is divisible by $\hat{n}$, the outage probability in the FH system can be written as

$$\begin{aligned} \Pr\{\mathcal{O}_{i,\text{FH}}(R)\} &= \Pr\{\mathcal{O}_{i,\text{FH}}(R) | \mathfrak{N} \leq \hat{n}\} \Pr\{\mathfrak{N} \leq \hat{n}\} \\ &\quad + \Pr\{\mathcal{O}_{i,\text{FH}}(R) | \mathfrak{N} > \hat{n}\} \Pr\{\mathfrak{N} > \hat{n}\} \\ &\leq \Pr\{\mathcal{O}_{i,\text{FH}}(R) | \mathfrak{N} \leq \hat{n}\} + \Pr\{\mathfrak{N} > \hat{n}\} \\ &\leq \Pr\{\mathcal{O}_{i,\text{FH}}(R) | \mathfrak{N} = \hat{n}\} + \Pr\{\mathfrak{N} > \hat{n}\}. \end{aligned} \quad (113)$$

The last line follows from the fact that for a fixed R the outage probability is an increasing function of the number of active users. Choosing $v = \frac{u}{\hat{n}}$ in (100) and selecting the transmission rate as

$$\hat{R} = \frac{u}{\hat{n}} \log\left(1 - \kappa_{\frac{u}{\hat{n}}} \gamma^{(1 - \frac{1}{\hat{n}})^{\hat{n}-1}} \ln\left(1 - \frac{\varepsilon}{2}\right)\right) \quad (114)$$

results in $\Pr\{\mathcal{O}_{i,\text{FH}}(\hat{R}) | \mathfrak{N} = \hat{n}\} \leq \frac{\varepsilon}{2}$ where $\kappa_{\frac{u}{\hat{n}}} = \frac{2^{-\mathfrak{H}(\frac{u}{\hat{n}}, \hat{n})}}{\frac{u}{\hat{n}}} \left(\frac{\hat{n}(\hat{n}-1)}{u}\right)^{(1 - \frac{1}{\hat{n}})^{\hat{n}-1} - 1}$. Furthermore, we select $\hat{n}$ such that

$$\Pr\{\mathfrak{N} > \hat{n}\} \leq \frac{\varepsilon}{2}. \quad (115)$$

Noting that

$$\begin{aligned} \Pr\{\mathfrak{N} > \hat{n}\} &= \Pr\{\tilde{\mathfrak{N}} \geq \hat{n}\} \\ &\approx \sum_{n=\hat{n}}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \end{aligned} \quad (116)$$

¹³We note that $\tilde{\mathfrak{N}} = \mathfrak{N} - 1$.

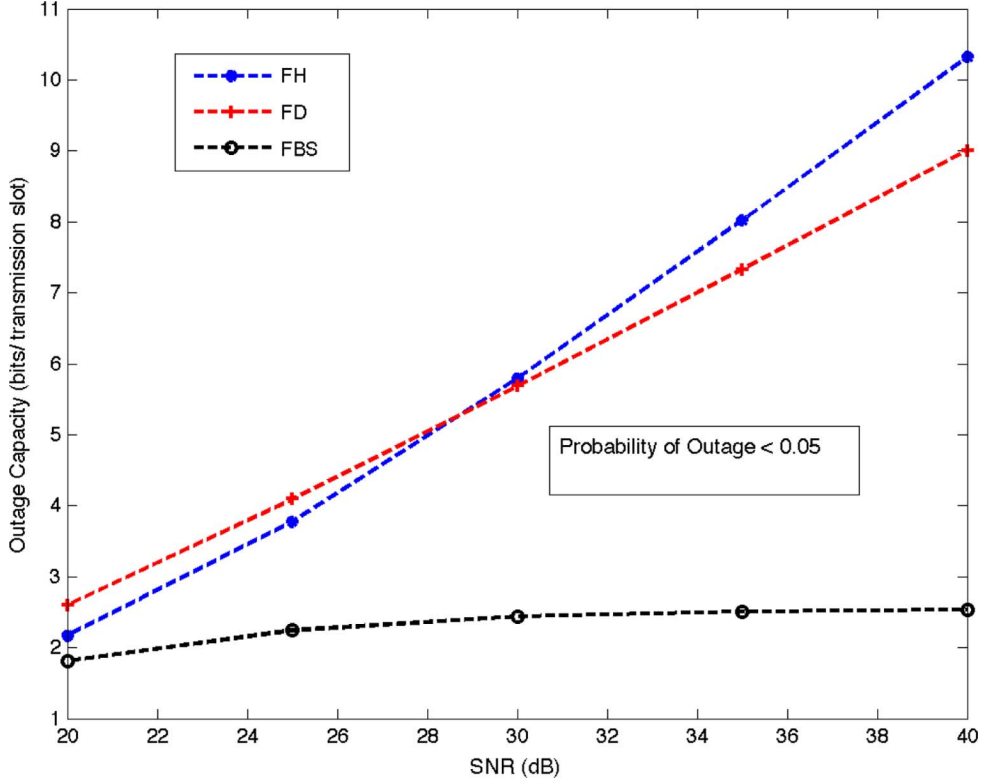


Fig. 8. Comparison of FH, FD, and FBS in terms of outage capacity (bits/transmission slot) for $u = n_{\max} = 10$ and $\{q_n\}_{n=1}^{n_{\max}}$ given in (119). The probability of outage is not allowed to exceed 0.05.

it is shown in Appendix F that for small enough ε , selecting $\hat{n} = -\lambda \ln \varepsilon$ guarantees $\Pr\{\mathfrak{N} > \hat{n}\} \leq \frac{\varepsilon}{2}$. Hence, using (114)

$$\begin{aligned}
 R_{\text{FH}}(\varepsilon) &\stackrel{(a)}{\geq} \hat{R} \\
 &\geq \frac{u}{\hat{n}} \log \left(-\kappa_{\frac{u}{\hat{n}}} \gamma^{(1-\frac{1}{\hat{n}})^{\hat{n}-1}} \ln \left(1 - \frac{\varepsilon}{2} \right) \right) \\
 &= \frac{u}{\hat{n}} \left(1 - \frac{1}{\hat{n}} \right)^{\hat{n}-1} \log \gamma \\
 &\quad + \frac{u}{\hat{n}} \log \left(-\kappa_{\frac{u}{\hat{n}}} \ln \left(1 - \frac{\varepsilon}{2} \right) \right) \\
 &\approx \frac{u}{\hat{n}} \left(1 - \frac{1}{\hat{n}} \right)^{\hat{n}-1} \log \gamma \\
 &\stackrel{(b)}{\geq} -\frac{u}{e\lambda \ln \varepsilon} \log \gamma
 \end{aligned} \tag{117}$$

as $\gamma \rightarrow \infty$. In (117), (a) follows by the fact that setting the transmission rate at $\hat{R}$, we get $\Pr\{\mathcal{O}_{i,\text{FH}}(\hat{R})\} \leq \varepsilon$. This can be easily seen by (113) for the particular choice of $\hat{n} = -\lambda \ln \varepsilon$. Also, (b) holds since $(1 - \frac{1}{\hat{n}})^{\hat{n}-1}$ is greater or equal to $\frac{1}{e}$.

By (111) and setting $n_{\text{des}} = u$,

$$R_{\text{FD}}(\varepsilon) \approx \log \gamma. \tag{118}$$

The above equations imply that as long as $u > -e\lambda \ln \varepsilon$, FH is superior to FD. This condition can be alternatively written as $p < -\frac{1}{e \ln \varepsilon}$.

E. Numerical Results

We consider a practical scenario where each user operates in two different environments with distinct distributions on the number of active users. To describe the distribution of $\mathfrak{N}$, let $\mathfrak{B}_1$ and $\mathfrak{B}_2$ be two independent Binomial random variables with parameters (n_1, p_1) and (n_2, p_2) , respectively. For some $p_0 \in (0, 1)$, we set

$$\begin{aligned}
 q_n &= p_0 \Pr\{\mathfrak{B}_1 = n | \mathfrak{B}_1 \geq 1\} \\
 &\quad + (1 - p_0) \Pr\{\mathfrak{B}_2 = n | \mathfrak{B}_2 \geq 1\} \\
 &= p_0 \frac{\binom{n_1}{n} p_1^n (1 - p_1)^{n_1-n}}{1 - (1 - p_1)^{n_1}} \\
 &\quad + (1 - p_0) \frac{\binom{n_2}{n} p_2^n (1 - p_2)^{n_2-n}}{1 - (1 - p_2)^{n_2}}
 \end{aligned} \tag{119}$$

where $1 \leq n \leq \max\{n_1, n_2\}$. In fact, the number of active users is a Binomial random variable with parameters (n_1, p_1) for a fraction p_0 of the time and a Binomial random variable with parameters (n_2, p_2) for the rest of the time. We have made the assumption that there is always at least one active user in the network.

Let us set $n_1 = 3$, $n_2 = 10$, $p_1 = 0.1$, $p_2 = 0.2$, and $p_0 = 0.8$. Therefore, $n_{\max} = \max\{n_1, n_2\} = 10$. We assume there are $u = 10$ frequency subbands. Fig. 8 depicts $\max_v R_{\text{FH}}^{(2)}(\varepsilon)$, $R_{\text{FD}}(\varepsilon)$ and $R_{\text{FBS}}(\varepsilon)$ in terms of SNR for $\varepsilon = 0.05$. It is seen that for $\gamma > 29$ dB the FH scenario offers a better outage capacity compared to the FD scheme. It is also observed that the FBS scheme offers a poor performance compared to both FD and FH.

VIII. CONCLUSION

We have addressed a decentralized wireless communication network with a fixed number u of frequency subbands to be shared among N transmitter-receiver pairs. It is assumed that the number of active users is a realization of a random variable with a given distribution. Moreover, users are assumed to be unaware of each other's codebooks and hence, no multiuser detection is possible. We proposed a randomized FH scheme in which each transmitter randomly hops over subsets of the u subbands from transmission slot to transmission slot. Assuming all users transmit Gaussian signals, the distribution of noise plus interference is mixed Gaussian, which makes the calculation of the mutual information between the input and output of each user intractable. We derived lower and upper bounds on this mutual information and demonstrated that for large SNR values, the two bounds coincide. This observation enabled us to compute the sum multiplexing gain of the system and obtain the optimum hopping strategy for maximizing this value. We compared the performance of the FH with that of the FD in terms of the following performance measures: average sum multiplexing gain ($\eta^{(1)}$) and average minimum multiplexing gain per user ($\eta^{(2)}$). We showed that (depending on the probability mass function of the number of active users) the FH system can offer a significant improvement in terms of $\eta^{(1)}$ and $\eta^{(2)}$ (implying a more efficient usage of the spectrum). In the sequel, we considered a scenario where the channel gains are quasi-static Rayleigh fading. The transmitters are assumed to be unaware of the number of active users in the network as well as the channel gains. Developing a new upper bound on the differential entropy of a mixed Gaussian random vector and via entropy power inequality, we offered three lower bounds on the ε -outage capacity for each user in the proposed scheme. Asymptotic analysis was presented in terms of SNR and outage threshold. In the asymptotically small ε regime, we observed that the maximum outage capacity is obtained for either $v = 1$ or $v = u$; in the asymptotically small SNR regime, we demonstrated that for all values of v the system achieves the optimal performance; for asymptotically large SNR, it is shown that $v_{\text{opt}} = \left\lceil \frac{u}{n_{\text{max}}} \right\rceil$, where n_{max} is the maximum number of concurrently active users in the network. We compared the outage capacity of the underlying FH scheme with that of the FD scenario for various setups in terms of distributions on the number of active users, SNR and ε and showed that FH outperforms FD in many cases.

APPENDIX A

PROOF OF LEMMAS 1 AND 2

Let us consider a $t \times 1$ complex vector $\vec{\Theta}$ with mixed Gaussian distribution $p_{\vec{\Theta}}(\vec{\theta})$ with different covariance matrices $\{C_l\}_{l=1}^L$ and associated probabilities $\{p_l\}_{l=1}^L$ given by

$$p_{\vec{\Theta}}(\vec{\theta}) = \sum_{l=1}^L p_l g_t^{(c)}(\vec{\theta}, C_l) \quad (120)$$

where $g_t^{(c)}(\vec{\theta}, C_l) = \frac{1}{\pi^t \det C_l} \exp\left(-\vec{\theta}^H C_l^{-1} \vec{\theta}\right)$ and it is assumed that $C_l = \varrho_l^2 I_t$ and $\varrho_1^2 < \varrho_2^2 < \dots < \varrho_L^2$. One can write

$$\int p_{\vec{\Theta}}(\vec{\theta}) \log p_{\vec{\Theta}}(\vec{\theta}) d\vec{\theta} = \sum_{l=1}^L J_l \quad (121)$$

where $J_l = p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log p_{\vec{\Theta}}(\vec{\theta}) d\vec{\theta}$ for $1 \leq l \leq L$. To find a proper lower bound on each J_l , we proceed as follows. We have

$$J_l = p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log \left(\sum_{m=1}^L p_m g_t^{(c)}(\vec{\theta}, C_m) \right) d\vec{\theta}. \quad (122)$$

On the other hand, one can write $\log \left(\sum_{m=1}^L p_m g_t^{(c)}(\vec{\theta}, C_m) \right)$ as in (123) at the bottom of the page. However, the term $\sum_{m=l+1}^L \frac{p_m g_t^{(c)}(\vec{\theta}, C_m)}{p_l g_t^{(c)}(\vec{\theta}, C_l)} = \sum_{m=l+1}^L \frac{p_m \varrho_l^{2t}}{p_l \varrho_m^{2t}} \exp - \left(\left(\frac{1}{\varrho_m^2} - \frac{1}{\varrho_l^2} \right) \vec{\theta}^H \vec{\theta} \right)$ is always greater than $\sum_{m=l+1}^L \frac{p_m \varrho_l^{2t}}{p_l \varrho_m^{2t}}$. Hence, we arrive at (124) at the bottom of the page. On the other hand, the term $\sum_{m=1}^{l-1} \frac{p_m g_t^{(c)}(\vec{\theta}, C_m)}{p_l g_t^{(c)}(\vec{\theta}, C_l)} = \sum_{m=1}^{l-1} \frac{p_m \varrho_l^{2t}}{p_l \varrho_m^{2t}} \exp - \left(\left(\frac{1}{\varrho_m^2} - \frac{1}{\varrho_l^2} \right) \vec{\theta}^H \vec{\theta} \right)$ is always less than $\sum_{m=1}^{l-1} \frac{p_m \varrho_l^{2t}}{p_l \varrho_m^{2t}}$. Now, we use the following inequality,¹⁴ which is valid for any $b > 0$ and $0 \leq x \leq a$

$$\log(1+b+x) \geq \left(1 - \frac{x}{a}\right) \log(1+b) + \frac{x}{a} \log(1+a+b). \quad (125)$$

¹⁴One may verify this using Jensen's inequality and concavity of the $\log(\cdot)$ function.

$$\log \left(\sum_{m=1}^L p_m g_t^{(c)}(\vec{\theta}, C_m) \right) = \log \left(p_l g_t^{(c)}(\vec{\theta}, C_l) \right) + \log \left(1 + \sum_{m=1}^{l-1} \frac{p_m g_t^{(c)}(\vec{\theta}, C_m)}{p_l g_t^{(c)}(\vec{\theta}, C_l)} + \sum_{m=l+1}^L \frac{p_m g_t^{(c)}(\vec{\theta}, C_m)}{p_l g_t^{(c)}(\vec{\theta}, C_l)} \right). \quad (123)$$

$$\log \left(\sum_{m=1}^L p_m g_t^{(c)}(\vec{\theta}, C_m) \right) \geq \log \left(p_l g_t^{(c)}(\vec{\theta}, C_l) \right) + \log \left(1 + \sum_{m=l+1}^L \frac{p_m \varrho_l^{2t}}{p_l \varrho_m^{2t}} + \sum_{m=1}^{l-1} \frac{p_m g_t^{(c)}(\vec{\theta}, C_m)}{p_l g_t^{(c)}(\vec{\theta}, C_l)} \right). \quad (124)$$

Utilizing this in the expression on the RHS of (124), we get (126) at the bottom of the page where $\mu_l = \sum_{m=l+1}^L \frac{p_m}{p_l} \frac{\varrho_l^{2t}}{\varrho_m^{2t}}$ and $\nu_l = \sum_{m=1}^{l-1} \frac{p_m}{p_l} \frac{\varrho_l^{2t}}{\varrho_m^{2t}}$. Using this in (122) yields

$$\begin{aligned} J_l &\geq p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log(p_l g_t^{(c)}(\vec{\theta}, C_l)) d\vec{\theta} \\ &\quad + \left(p_l - \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \right) \log(1 + \mu_l) \\ &\quad + \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \log(1 + \nu_l + \mu_l). \end{aligned} \quad (127)$$

The first term on the RHS can be calculated as

$$\begin{aligned} p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log(p_l g_t^{(c)}(\vec{\theta}, C_l)) d\vec{\theta} &= p_l \log(p_l) \int g_t^{(c)}(\vec{\theta}, C_l) d\vec{\theta} \\ &\quad + p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log g_t^{(c)}(\vec{\theta}, C_l) d\vec{\theta} \\ &= p_l \log p_l + p_l \int g_t^{(c)}(\vec{\theta}, C_l) \log g_t^{(c)}(\vec{\theta}, C_l) d\vec{\theta} \\ &\stackrel{(a)}{=} p_l \log p_l - p_l \log((\pi e)^t \det C_l) \\ &= p_l \log p_l - t p_l \log(\pi e \varrho_l^2) \end{aligned} \quad (128)$$

where in (a) we have used the fact that the differential entropy of a $t \times 1$ complex Gaussian vector with covariance matrix C_l is $\log((\pi e)^t \det C_l)$. Thus

$$\begin{aligned} h(\vec{\Theta}) &= - \int p_{\vec{\Theta}}(\vec{\theta}) \log p_{\vec{\Theta}}(\vec{\theta}) d\vec{\theta} = - \sum_{l=1}^L J_l \\ &\leq - \sum_{l=1}^L p_l \log p_l + t \sum_{l=1}^L p_l \log(\pi e \varrho_l^2) \\ &\quad - \sum_{l=1}^L \left(p_l - \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \right) \log(1 + \mu_l) \\ &\quad - \sum_{l=1}^L \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \log(1 + \nu_l + \mu_l). \end{aligned} \quad (129)$$

Briefly

$$h(\vec{Z}) \leq t \sum_{l=1}^L p_l \log(\pi e \varrho_l^2) + \mathfrak{H} - \mathfrak{G}'' \quad (130)$$

where

$$\mathfrak{H} = - \sum_{l=1}^L p_l \log p_l \quad (131)$$

and

$$\begin{aligned} \mathfrak{G}'' &= \sum_{l=1}^L \left(p_l - \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \right) \log(1 + \mu_l) \\ &\quad + \sum_{l=1}^L \frac{\sum_{m=1}^{l-1} p_m}{\nu_l} \log(1 + \nu_l + \mu_l). \end{aligned} \quad (132)$$

$\mathfrak{G}''$ is a complicated function of $\{\varrho_l\}_{l=1}^L$. To simplify it, one may notice that $\mathfrak{G}''$ is an increasing function of μ_l . Hence, using $\mu_l \geq 0$, we get a lower bound on $\mathfrak{G}''$, namely $\mathfrak{G}'$ given by

$$\mathfrak{G}' = \sum_{l=2}^L \frac{\log(1 + \nu_l)}{\nu_l} \sum_{m=1}^{l-1} p_m. \quad (133)$$

On the other hand, using the fact that $\frac{\log(1+x)}{x}$ is a decreasing function of x , one may obtain a lower bound on $\mathfrak{G}'$ by finding an upper bound on ν_l for each l . One option is $\nu_l \leq \frac{\varrho_l^{2t}}{\varrho_1^{2t}} \frac{\sum_{m=1}^{l-1} p_m}{p_l}$. Thus, we come up with the following lower bound on $\mathfrak{G}'$

$$\begin{aligned} \mathfrak{G}' &\geq \mathfrak{G} \\ &\triangleq \frac{\varrho_1^{2t}}{\varrho_L^{2t}} \sum_{l=2}^L p_l \log \left(1 + \frac{\varrho_L^{2t}}{\varrho_1^{2t}} \frac{\sum_{m=1}^{l-1} p_m}{p_l} \right). \end{aligned} \quad (134)$$

APPENDIX B PROOF OF (42)

Let us consider the generalized FH scheme with hopping parameter $v = v^*$. The number of frequency subbands chosen by the i th user in a typical transmission slot is denoted by the random variable V_i . This number changes independently from transmission slot to transmission slot. In fact, $\Pr\{V_i = \lfloor v^* \rfloor\} = \lceil v^* \rceil - v^*$ and $\Pr\{V_i = \lceil v^* \rceil\} = v^* - \lfloor v^* \rfloor$. We mention that $\{V_j\}_{j=1}^N$ are independent random variables. As before, we denote the set of frequency subbands selected by the i th user by S_i where $|S_i| = V_i$. To be more clear, we use the notation $S_i(V_i)$ instead of S_i . The achievable rate of the i th user is given by (135) at the bottom of the page where $s_i^*(v_i)$ denotes the selec-

$$\begin{aligned} \log \left(\sum_{m=1}^L p_m g_t^{(c)}(\vec{\theta}, C_m) \right) &\geq \left(1 - \frac{1}{\nu_l} \sum_{m=1}^{l-1} \frac{p_m}{p_l} \frac{g_t^{(c)}(\vec{\theta}, C_m)}{g_t^{(c)}(\vec{\theta}, C_l)} \right) \log(1 + \mu_l) \\ &\quad + \frac{1}{\nu_l} \sum_{m=1}^{l-1} \frac{p_m}{p_l} \frac{g_t^{(c)}(\vec{\theta}, C_m)}{g_t^{(c)}(\vec{\theta}, C_l)} \log(1 + \nu_l + \mu_l) \end{aligned} \quad (126)$$

$$\mathcal{R}_i = I(\vec{X}_i; \vec{Y}_i | V_i, S_i(V_i)) = \sum_{v_i \in \{\lfloor v^* \rfloor, \lceil v^* \rceil\}} \Pr\{V_i = v_i\} I(\vec{X}_i; \vec{Y}_i | V_i = v_i, S_i(V_i) = s_i^*(v_i)) \quad (135)$$

tion of the first v_i subbands by the i th user. Following the same lines that led to the lower bound in Section III,

$$I(\vec{X}_i; \vec{Y}_i | V_i = v_i, S_i(V_i) = s_i^*(v_i)) \geq \frac{v_i}{2} \log \left(2^{\frac{2}{v_i} (h(\vec{X}_i(s_i^*(v_i))) - h(\vec{Z}'_i))} \right) \quad (136)$$

where $\vec{Z}'_i$ consists of the first v_i elements of $\vec{Z}_i$. On the other hand

$$\begin{aligned} h(\vec{Z}'_i) &\leq h(\vec{Z}'_i, \{V_j\}_{j \neq i}) \\ &= h(\vec{Z}'_i | \{V_j\}_{j \neq i}) + H(\{V_j\}_{j \neq i}) \\ &= h(\vec{Z}'_i | \{V_j\}_{j \neq i}) + \sum_{j \neq i} H(V_j) \\ &= h(\vec{Z}'_i | \{V_j\}_{j \neq i}) \\ &\quad + (N-1)\mathcal{H}(\lceil v^* \rceil - v^*). \end{aligned} \quad (137)$$

Let us define

$$\tilde{V}_j = \begin{cases} V_j & 1 \leq j \leq i-1 \\ V_{j+1} & i \leq j \leq N-1 \end{cases}. \quad (138)$$

Then, we have the thread in (139) at the bottom of the page. Using Lemma 1

$$\begin{aligned} h(Z_{i,1} | \tilde{V}_1 = \tilde{v}_1, \dots, \tilde{V}_{N-1} = \tilde{v}_{N-1}) \\ \leq \frac{1 - a_{i,0}(\tilde{v}_1, \dots, \tilde{v}_{N-1})}{2} \log(c_{i,L_i}(\tilde{v}_1, \dots, \tilde{v}_{N-1})\gamma + 1) \\ - \sum_{l=0}^{L_i} a_{i,l}(\tilde{v}_1, \dots, \tilde{v}_{N-1}) \log a_{i,l}(\tilde{v}_1, \dots, \tilde{v}_{N-1}) \\ + \log(\sqrt{2\pi e}\sigma). \end{aligned} \quad (140)$$

Assuming the channel gains are realizations of independent and continuous random variables, then $L_i = 2^{N-1} - 1$. The set $\{a_{i,l}(\tilde{v}_1, \dots, \tilde{v}_{N-1})\}_{l=0}^{2^{N-1}-1}$ consists of the numbers $\prod_{k=1}^n \frac{\tilde{v}_{m_k}}{u} \prod_{\substack{1 \leq m \leq N-1 \\ m \notin \{m_k\}_{k=1}^n}} \left(1 - \frac{\tilde{v}_m}{u}\right)$ for $1 \leq n \leq N-1$ and $1 \leq m_1 < \dots < m_n \leq N-1$. Also,

$$a_{i,0}(\tilde{v}_1, \dots, \tilde{v}_{N-1}) = \prod_{j=1}^{N-1} \left(1 - \frac{\tilde{v}_j}{u}\right) \quad (141)$$

and

$$c_{i,L_i}(\tilde{v}_1, \dots, \tilde{v}_{N-1}) = \sum_{j \neq i} \frac{|h_{j,i}|^2}{\tilde{v}_j}. \quad (142)$$

Clearly,

$$c_{i,L_i}(\tilde{v}_1, \dots, \tilde{v}_{N-1}) \leq \frac{\sum_{j \neq i} |h_{j,i}|^2}{\lfloor v^* \rfloor}. \quad (143)$$

The (139) to (143) yield (144) at the bottom of the page. It is straightforward to show that the last term in (144) is equal to $(N-1)v_i\vartheta_{v^*}$ where ϑ_{v^*} is given in (43). Finally, by (135), (136), (137), and (144)

$$\mathcal{R}_i \geq E \left\{ \frac{V_i}{2} \log \left(\frac{2^{-2(N-1)} \left(\vartheta_{v^*} + \frac{\mathcal{H}(\lceil v^* \rceil - v^*)}{V_i} \right) \gamma |h_{i,i}|^2}{V_i \left(\frac{\gamma \sum_{j \neq i} |h_{j,i}|^2}{\lfloor v^* \rfloor} + 1 \right)^{1 - \left(1 - \frac{v^*}{u}\right)^{N-1}}} + 1 \right) \right\}. \quad (145)$$

$$\begin{aligned} h(\vec{Z}'_i | \{V_j\}_{j \neq i}) &= h(\vec{Z}'_i | \{\tilde{V}_j\}_{j=1}^{N-1}) \\ &= \sum_{\tilde{v}_1, \dots, \tilde{v}_{N-1} \in \{\lceil v^* \rceil, \lfloor v^* \rfloor\}} \prod_{j=1}^{N-1} \Pr\{\tilde{V}_j = \tilde{v}_j\} h(\vec{Z}'_i | \tilde{V}_1 = \tilde{v}_1, \dots, \tilde{V}_{N-1} = \tilde{v}_{N-1}) \\ &\leq v_i \sum_{\tilde{v}_1, \dots, \tilde{v}_{N-1} \in \{\lceil v^* \rceil, \lfloor v^* \rfloor\}} \prod_{j=1}^{N-1} \Pr\{\tilde{V}_j = \tilde{v}_j\} h(Z_{i,1} | \tilde{V}_1 = \tilde{v}_1, \dots, \tilde{V}_{N-1} = \tilde{v}_{N-1}). \end{aligned} \quad (139)$$

$$\begin{aligned} h(\vec{Z}_i | \{V_j\}_{j \neq i}) &\leq \frac{v_i}{2} \left(1 - \prod_{j=1}^{N-1} \left(1 - \frac{E\{\tilde{V}_j\}}{u} \right) \right) \log \left(\frac{\gamma \sum_{j \neq i} |h_{j,i}|^2}{\lfloor v^* \rfloor} + 1 \right) + v_i \log(\sqrt{2\pi e}\sigma) \\ &\quad - v_i \sum_{n=0}^{N-1} \sum_{\{m_k\}_{k=1}^n} E \left\{ \prod_{k=1}^n \frac{\tilde{V}_{m_k}}{u} \prod_{\substack{1 \leq m \leq N-1 \\ m \notin \{m_k\}_{k=1}^n}} \left(1 - \frac{\tilde{V}_m}{u} \right) \log \left(\prod_{k=1}^n \frac{\tilde{V}_{m_k}}{u} \prod_{\substack{1 \leq m \leq N-1 \\ m \notin \{m_k\}_{k=1}^n}} \left(1 - \frac{\tilde{V}_m}{u} \right) \right) \right\}. \end{aligned} \quad (144)$$

APPENDIX C
PROOF OF PROPOSITION 1

We have that $\eta_{\text{FD}}^{(1)} = \frac{E\{\mathfrak{N}\}u}{2n_{\text{max}}}$ and $\eta_{\text{FH}}^{(1)} = \frac{1}{2} \max_v \left\{ vE \left\{ \mathfrak{N} \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1} \right\} \right\}$. Let $\Omega(v, \mathfrak{N}) \triangleq \mathfrak{N}\omega_v^{\mathfrak{N}-1}$ where $\omega_v = 1 - \frac{v}{u}$. Thinking of $\mathfrak{N}$ as a real parameter for the moment, we have $\frac{\partial^2}{\partial \mathfrak{N}^2} \Omega(v, \mathfrak{N}) = \omega_v^{\mathfrak{N}-1} \left((\ln \omega_v)^2 + 2 \ln \omega_v \right)$. As $\mathfrak{N} \geq 1$, then $\frac{\partial^2}{\partial \mathfrak{N}^2} \Omega(v, \mathfrak{N}) \geq \omega_v^{\mathfrak{N}-1} \left((\ln \omega_v)^2 + 2 \ln \omega_v \right)$. But, $(\ln \omega_v)^2 + 2 \ln \omega_v \geq 0$ if and only if $\omega_v \leq \frac{1}{e^2}$ or $\omega_v \geq 1$. Since $\omega_v \leq 1$, we get $\omega_v \leq \frac{1}{e^2}$. This implies that the function $\Omega(v, \mathfrak{N})$ is a convex function of $\mathfrak{N}$ as far as $\omega_v \leq \frac{1}{e^2}$. Therefore, by Jensen's inequality

$$\begin{aligned} E \left\{ \mathfrak{N} \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1} \right\} &= E\{\Omega(v, \mathfrak{N})\} \\ &\geq \Omega(v, E\{\mathfrak{N}\}) \\ &= E\{\mathfrak{N}\} \left(1 - \frac{v}{u}\right)^{E\{\mathfrak{N}\}-1} \end{aligned} \quad (146)$$

which is valid as far as $v \geq (1 - \frac{1}{e^2})u$. Hence

$$\begin{aligned} \eta_{\text{FH}}^{(1)} &= \frac{1}{2} \max_v \left\{ vE \left\{ \mathfrak{N} \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1} \right\} \right\} \\ &\geq \frac{1}{2} \max_{v \in [(1-\frac{1}{e^2})u, u]} \left\{ vE \left\{ \mathfrak{N} \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1} \right\} \right\} \\ &\geq \frac{1}{2} E\{\mathfrak{N}\} \max_{v \in [(1-\frac{1}{e^2})u, u]} \left\{ v \left(1 - \frac{v}{u}\right)^{E\{\mathfrak{N}\}-1} \right\}. \end{aligned} \quad (147)$$

The function $v \left(1 - \frac{v}{u}\right)^{E\{\mathfrak{N}\}-1}$ is a concave function in terms of v that achieves its absolute maximum at $\frac{u}{E\{\mathfrak{N}\}}$. Therefore

$$\max_{v \in [(1-\frac{1}{e^2})u, u]} \left\{ v \left(1 - \frac{v}{u}\right)^{E\{\mathfrak{N}\}-1} \right\} = \mathcal{Q}(1 - \mathcal{Q})^{E\{\mathfrak{N}\}-1} u \quad (148)$$

where $\mathcal{Q} \triangleq \max \left\{ 1 - \frac{1}{e^2}, \frac{1}{E\{\mathfrak{N}\}} \right\}$. Using (147) and (148)

$$\eta_{\text{FH}}^{(1)} \geq \frac{1}{2} \mathcal{Q}(1 - \mathcal{Q})^{E\{\mathfrak{N}\}-1} E\{\mathfrak{N}\}u. \quad (149)$$

Hence, a sufficient condition for $\eta_{\text{FH}}^{(1)} > \eta_{\text{FD}}^{(1)}$ to hold is that

$$\mathcal{Q}(1 - \mathcal{Q})^{E\{\mathfrak{N}\}-1} > \frac{1}{n_{\text{max}}}. \quad (150)$$

If $E\{\mathfrak{N}\} \geq \frac{e^2}{e^2-1}$, we have $\mathcal{Q} = 1 - \frac{1}{e^2}$. Hence, (150) reduces to the inequality $E\{\mathfrak{N}\} < \frac{1}{2} \ln((e^2-1)n_{\text{max}})$. Therefore, if $\frac{e^2}{e^2-1} \leq E\{\mathfrak{N}\} < \frac{1}{2} \ln((e^2-1)n_{\text{max}})$, then (150) is satisfied. On the other hand, if $E\{\mathfrak{N}\} \leq \frac{e^2}{e^2-1} = 1.1565$, we get $\mathcal{Q} = \frac{1}{E\{\mathfrak{N}\}}$. Thus, (150) re-

duces to $\frac{1}{E\{\mathfrak{N}\}} \left(1 - \frac{1}{E\{\mathfrak{N}\}}\right)^{E\{\mathfrak{N}\}-1} > \frac{1}{n_{\text{max}}}$. For each $n_{\text{max}} \geq 2$, this yields an upper bound on $E\{\mathfrak{N}\}$. Since $\frac{1}{E\{\mathfrak{N}\}} \left(1 - \frac{1}{E\{\mathfrak{N}\}}\right)^{E\{\mathfrak{N}\}-1}$ is a decreasing function of $E\{\mathfrak{N}\}$, the smallest of these upper bounds is obtained for $n_{\text{max}} = 2$ and is equal to 1.2938. This means that for $E\{\mathfrak{N}\} \leq 1.1565$, (150) is automatically satisfied. Thus, (150) is equivalent to

$$E\{\mathfrak{N}\} < \frac{1}{2} \ln((e^2-1)n_{\text{max}}). \quad (151)$$

To prove the second part, we note that $\eta_{\text{FD}}^{(2)} = \frac{u}{2n_{\text{max}}}$ and $\eta_{\text{FH}}^{(2)} = \frac{1}{2} \max_v \left\{ vE \left\{ \left(1 - \frac{v}{u}\right)^{N-1} \right\} \right\}$. The function $\left(1 - \frac{v}{u}\right)^{N-1}$ is convex in terms of N . Using Jensen's inequality,

$$\begin{aligned} \eta_{\text{FH}}^{(2)} &\geq \frac{1}{2} \max_v \left\{ v \left(1 - \frac{v}{u}\right)^{E\{N\}-1} \right\} \\ &= \frac{u}{2E\{N\}} \left(1 - \frac{1}{E\{N\}}\right)^{E\{N\}-1}. \end{aligned} \quad (152)$$

Hence, a sufficient condition for $\eta_{\text{FH}}^{(2)} > \eta_{\text{FD}}^{(2)}$ to hold is

$$\frac{1}{E\{N\}} \left(1 - \frac{1}{E\{N\}}\right)^{E\{N\}-1} > \frac{1}{n_{\text{max}}}. \quad (153)$$

APPENDIX D
COMPUTATION OF $R_{\text{FH}}^{(1)}(\varepsilon)$

By (66), we get (154) at the bottom of the page where $\beta_{m,\mathfrak{N}}\left(\frac{v}{u}\right) = \left(\frac{v}{u}\right)^m \left(1 - \frac{v}{u}\right)^{\mathfrak{N}-1-m}$ and for each m , $\{\mathcal{I}_{m,m'}\}_{m'=1}^{\binom{m}{\mathfrak{N}-1}}$ consists of all possible summations of m elements in the set $\{|\mathfrak{h}_{j,i}|^2\}_{j=1, j \neq i}^{\mathfrak{N}}$, such that $\prod_{m=1}^{\mathfrak{N}-1} \prod_{m'=1}^{\binom{m}{\mathfrak{N}-1}} \left(\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1\right)^{\beta_{m,\mathfrak{N}}\left(\frac{v}{u}\right)} = \prod_{l=1}^{\mathfrak{L}_i} (\mathbf{c}_{i,l} \gamma + 1)^{\mathbf{a}_{i,l}}$. We have also substituted $\mathfrak{G}_i$ by $\mathfrak{G}_{i,\text{lb}}(v, \mathfrak{N}) = \alpha_{\mathfrak{N}}(\mathcal{I}_{\mathfrak{N}-1,1}; \frac{\gamma}{v}, \frac{v}{u})$. Since $\mathfrak{N}$ itself is a random variable, the outage probability can be written as

$$\Pr \left\{ \mathcal{R}_{i,\text{lb}}^{(1)}(\vec{\mathfrak{h}}_i) < R \right\} = q_1 \xi_1 + \sum_{n=2}^{n_{\text{max}}} q_n \xi_n \quad (155)$$

where

$$\begin{aligned} \xi_1 &= \Pr \left\{ \mathcal{R}_{i,\text{lb}}^{(1)}(\vec{\mathfrak{h}}_i) < R | \mathfrak{N} = 1 \right\} \\ &= \Pr \left\{ v \log \left(1 + \frac{|\mathfrak{h}_{i,i}|^2 \gamma}{v} \right) < R \right\} \\ &= 1 - \exp \left(- \frac{\left(1 - 2^{\frac{R}{v}}\right)v}{\gamma} \right). \end{aligned} \quad (156)$$

$$\Pr \left\{ \mathcal{R}_{i,\text{lb}}^{(1)}(\vec{\mathfrak{h}}_i) < R \right\} = \Pr \left\{ v \log \left(\frac{2^{-\mathfrak{H}(v, \mathfrak{N})} 2^{\alpha_{\mathfrak{N}}(\mathcal{I}_{\mathfrak{N}-1,1}; \frac{\gamma}{v}, \frac{v}{u})} |\mathfrak{h}_{i,i}|^2 \gamma}{v \prod_{m=1}^{\mathfrak{N}-1} \prod_{m'=1}^{\binom{m}{\mathfrak{N}-1}} \left(\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1\right)^{\beta_{m,\mathfrak{N}}\left(\frac{v}{u}\right)}} + 1 \right) < R \right\} \quad (154)$$

Denoting the collection of random variables $\{\mathcal{I}_{m,m'}\}_{1 \leq m \leq n-1, 1 \leq m' \leq \binom{n-1}{m}}$ by $\mathcal{I}_n$, the quantity ξ_n is calculated in (157) for $n \geq 2$ [see (157) at the bottom of the page] where (a) follows from the fact that after conditioning on $\mathcal{I}_n$, the only random quantity is $|\mathbf{h}_{i,i}|^2$, which is exponentially distributed. Also, (b) holds by the definition of ψ_n . This yields the desired result.

APPENDIX E

PROOF OF COROLLARY 1

As $b_1 < 0$ and $0 < 2^{-\alpha_n(\theta; b_2, p_2)} < 1$ for all $\theta > 0$, we have

$$\begin{aligned} \phi_n(b_{1,n}, b_2, p_{1,n}, p_2) &\geq \frac{1}{(n-2)!} \int_0^\infty \theta^{n-2} \exp(b_{1,n}(b_2\theta + 1)^{p_{1,n}} - \theta) d\theta. \end{aligned} \quad (158)$$

One may easily check that $\frac{\theta^{n-2}}{(n-2)!} \exp(-\theta) \mathbb{1}(\theta > 0)$ is a pdf for some nonnegative random variable Θ . Thus, (158) can be written as

$$\phi_n(b_{1,n}, b_2, p_{1,n}, p_2) \geq E\{\exp(b_{1,n}(b_2\Theta + 1)^{p_{1,n}})\}. \quad (159)$$

However, as $b_{1,n} < 0$ and $0 < p_{1,n} < 1$, the function $\exp(b_{1,n}(b_2\Theta + 1)^{p_{1,n}})$ is a convex function of Θ . Hence, applying Jensen's inequality yields

$$\begin{aligned} \phi_n(b_{1,n}, b_2, p_{1,n}, p_2) &\geq \exp(b_{1,n}(b_2 E\{\Theta\} + 1)^{p_{1,n}}) \\ &= \exp(b_{1,n}((n-1)b_2 + 1)^{p_{1,n}}), \end{aligned} \quad (160)$$

where we have used $E\{\Theta\} = n-1$. Using (160) in (82) and noting that $b_{1,n} = \frac{(1-2^{\frac{R}{v}})_v}{\gamma} 2^{\mathfrak{H}(v,n)}$, $b_2 = \frac{\gamma}{v}$, and $p_{1,n} = 1 - a(v,n)$, we get the desired lower bound.

APPENDIX F

Setting $\hat{n} = -\lambda \ln \varepsilon$, we have

$$\begin{aligned} \Pr\{\tilde{\mathfrak{N}} \geq \hat{n}\} &\approx \sum_{n=\hat{n}}^\infty \frac{e^{-\lambda} \lambda^n}{n!} \\ &= e^{-\lambda} \frac{\lambda^{\hat{n}}}{\hat{n}!} \left(1 + \sum_{j=1}^\infty \prod_{k=1}^j \frac{\lambda}{\hat{n}+k}\right) \\ &\leq e^{-\lambda} \frac{\lambda^{\hat{n}}}{\hat{n}!} \left[\sum_{j=0}^\infty \left(\frac{\lambda}{\hat{n}}\right)^j\right] \\ &= e^{-\lambda} \frac{\lambda^{\hat{n}}}{\hat{n}!} \frac{1}{1 - \frac{\lambda}{\hat{n}}} \\ &\stackrel{(a)}{\approx} \frac{e^{-\lambda}}{\sqrt{2\pi\hat{n}}} \left(\frac{\lambda e}{\hat{n}}\right)^{\hat{n}} \frac{1}{1 - \frac{\lambda}{\hat{n}}} \\ &= \frac{e^{-\lambda}}{\sqrt{2\pi\hat{n}}} \left(\frac{e}{-\ln \varepsilon}\right)^{\hat{n}} \frac{1}{1 + \frac{1}{\ln \varepsilon}} \end{aligned} \quad (161)$$

where (a) follows by stirling approximation for $\hat{n}!$. The term $\left(\frac{e}{-\ln \varepsilon}\right)^{\hat{n}}$ can be written as

$$\begin{aligned} \left(\frac{e}{-\ln \varepsilon}\right)^{\hat{n}} &= e^{-\hat{n}(\ln(-\ln \varepsilon) - 1)} \\ &= e^{\lambda(\ln(-\ln \varepsilon) - 1) \ln \varepsilon} \\ &\stackrel{(a)}{\leq} e^{\ln \varepsilon} \\ &= \varepsilon \end{aligned} \quad (162)$$

where (a) is valid if $\lambda(\ln(-\ln \varepsilon) - 1) \geq 1$, which is the case for sufficiently small ε . Combining this with the fact that $\frac{e^{-\lambda}}{\sqrt{2\pi\hat{n}}} \frac{1}{1 + \frac{1}{\ln \varepsilon}} < \frac{1}{2}$ gives the desired result.

$$\begin{aligned} \xi_n &= \Pr\{\mathcal{R}_{i,\text{lb}}^{(1)}(\vec{\mathbf{h}}_i) < R | \mathfrak{N} = n\} \\ &= \Pr\left\{v \log \left(\frac{2^{-\mathfrak{H}(v,n)} 2^{\alpha_n(\mathcal{I}_{n-1,1}; \frac{\gamma}{v}, \frac{v}{u})} |\mathbf{h}_{i,i}|^2 \gamma}{v \prod_{m=1}^{n-1} \prod_{m'=1}^{\binom{n-1}{m}} (\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1)^{\beta_{m,n}(\frac{v}{u})}} + 1 \right) < R \right\} \\ &= E\left\{ \Pr\left\{v \log \left(\frac{2^{-\mathfrak{H}(v,n)} 2^{\alpha_n(\mathcal{I}_{n-1,1}; \frac{\gamma}{v}, \frac{v}{u})} |\mathbf{h}_{i,i}|^2 \gamma}{v \prod_{m=1}^{n-1} \prod_{m'=1}^{\binom{n-1}{m}} (\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1)^{\beta_{m,n}(\frac{v}{u})}} + 1 \right) < R \middle| \mathcal{I}_n \right\} \right\} \\ &\stackrel{(a)}{=} E\left\{ 1 - \exp\left(-\frac{2^{\mathfrak{H}(v,n)} \left(1 - 2^{\frac{R}{v}}\right) v}{\gamma} 2^{-\alpha_n(\mathcal{I}_{n-1,1}; \frac{\gamma}{v}, \frac{v}{u})} \prod_{m=1}^{n-1} \prod_{m'=1}^{\binom{n-1}{m}} \left(\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1\right)^{\beta_{m,n}(\frac{v}{u})} \right) \right\} \\ &= 1 - E\left\{ \exp\left(-\frac{2^{\mathfrak{H}(v,n)} \left(1 - 2^{\frac{R}{v}}\right) v}{\gamma} 2^{-\alpha_n(\mathcal{I}_{n-1,1}; \frac{\gamma}{v}, \frac{v}{u})} \prod_{m=1}^{n-1} \prod_{m'=1}^{\binom{n-1}{m}} \left(\frac{\gamma}{v} \mathcal{I}_{m,m'} + 1\right)^{\beta_{m,n}(\frac{v}{u})} \right) \right\} \\ &\stackrel{(b)}{=} 1 - \psi_n\left(\frac{2^{\mathfrak{H}(v,n)} \left(1 - 2^{\frac{R}{v}}\right) v}{\gamma}, \frac{\gamma}{v}, \frac{v}{u} \right). \end{aligned} \quad (157)$$

ACKNOWLEDGMENT

The first author (K. Moshksar) is indebted to M. A. Maddah-Ali and S. Oveis Gharan for their invaluable suggestions. The authors are thankful to anonymous reviewers for their comments and suggestions that led to enrichment and refinement of the paper.

REFERENCES

- [1] Y. Liang, V. V. Veeravalli, and H. V. Poor, "Resource allocation for wireless fading relay channels: Max-min solution," *IEEE Trans. Inf. Theory*, vol. 53, no. 10, pp. 3432–3453, Oct. 2007.
- [2] K. Kumaran and H. Viswanathan, "Joint power and bandwidth allocation in downlink transmission," *IEEE Trans. Wireless Commun.*, vol. 4, no. 3, pp. 1008–1016, May 2005.
- [3] T. Holliday, A. Goldsmith, N. Bambos, and P. Glynn, "Distributed power and admission control for time-varying wireless networks," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT'04)*, Jul. 2004, pp. 352–352.
- [4] T. ElBatt and A. Ephremides, "Joint scheduling and power control for wireless ad hoc networks," *IEEE Trans. Wireless Commun.*, vol. 3, no. 1, pp. 74–85, Jan. 2004.
- [5] O. Seong-Jun, D. Zhang, and K. M. Wasserman, "Optimal resource allocation in multiservice CDMA networks," *IEEE Trans. Wireless Commun.*, vol. 2, no. 4, pp. 811–821, Jul. 2003.
- [6] Z. Han, Z. Ji, and K. J. R. Liu, "Fair multiuser channel allocation for OFDMA networks using Nash bargaining solutions and coalitions," *IEEE Trans. Commun.*, vol. 53, no. 8, pp. 1366–1376, Aug. 2005.
- [7] I. Katzela and M. Naghshineh, "Channel assignment schemes for cellular mobile telecommunication systems: A comprehensive survey," *IEEE Pers. Commun.*, vol. 3, no. 3, pp. 10–31, Jun. 1996.
- [8] S. G. Kiani and D. Gesbert, "Maximizing the capacity of large wireless networks: Optimal and distributed solutions," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT'06)*, Seattle, WA, Jul. 2006, pp. 2501–2505.
- [9] R. Etkin, A. P. Parekh, and D. Tse, "Spectrum sharing for unlicensed bands," *IEEE J. Sel. Areas Commun.*, vol. 25, pp. 517–528, Apr. 2007.
- [10] J. Mitola, "Cognitive radio: An integrated agent architecture for software defined radio," Ph.D., KTH, Stockholm, Sweden, 2000.
- [11] N. Devroye, P. Mitran, and V. Tarokh, "Achievable rates in cognitive radio channels," *IEEE Trans. Inf. Theory*, vol. 52, no. 5, pp. 1813–1827, May 2006.
- [12] A. Jovicic and P. Viswanath, "Cognitive radio: An information theoretic perspective," *IEEE Trans. Inf. Theory*, Apr. 2006, submitted for publication.
- [13] I. Maric, A. Goldsmith, G. Kramer, and S. Shamai(Shitz), "An achievable rate region for interference channels with a cognitive transmitter," in *Proc. 41st Asilomar Conf. Signals, Syst. Comput.*, Pacific Grove, CA, Nov. 2007.
- [14] S. A. Jafar and S. Srinivasa, "Capacity limits of cognitive radio with distributed and dynamic spectral activity," *IEEE J. Sel. Areas Commun.*, vol. 25, pp. 529–537, Apr. 2007.
- [15] A. Sahai, N. Hoven, S. M. Mishra, and R. Tandra, "Fundamental Trade-offs in Robust Spectrum Sensing for Opportunistic Frequency Reuse 2006 [Online]. Available: <http://www.eecs.berkeley.edu/sahai/papers/cognitivetechreport06.pdf>
- [16] C. M. Cordeiro and D. P. Agrawal, *Ad-hoc and Sensor Networks-Theory and Applications*. Singapore: World Sci., 2006.
- [17] J. Haartsen, "Bluetooth-The universal radio interface for ad-hoc wireless connectivity," *Ericsson Rev.*, no. 3, 1998.
- [18] T. Salonidis, P. Bhagwat, L. Tassiulas, and R. Lemaire, "Distributed topology construction of Bluetooth wireless personal area networks," *IEEE J. Sel. Areas Commun.*, vol. 23, pp. 633–643, Mar. 2005.
- [19] J. Haartsen and S. Mattisson, "Bluetooth- A new low power radio interface providing short range connectivity," *Proc. IEEE*, vol. 88, pp. 1651–1661, Oct. 2000.
- [20] Z. Kostic and N. Sollenberger, "Performance and implementation of dynamic frequency hopping in limited-bandwidth cellular systems," *IEEE Trans. Wireless Commun.*, vol. 1, no. 1, pp. 28–36, Jan. 2002.
- [21] O. Mubarek, H. Yanikomeroglu, and S. Periyalar, "Dynamic frequency hopping in cellular fixed relay networks," in *Proc. IEEE Veh. Technol. Conf.*, Jun. 2005, vol. 5, pp. 3112–3116.
- [22] Z. Kostic, I. Maric, and X. Wang, "Fundamentals of dynamic frequency hopping in cellular systems," *IEEE J. Sel. Areas Commun.*, vol. 19, no. 11, pp. 2254–2266, Nov. 2001.
- [23] R. Knopp and P. Humblet, "Information capacity and power control in single-cell multiuser communications," in *Proc. IEEE Int. Conf. Commun.*, Jun. 1995, vol. 1, pp. 331–335.
- [24] P. Viswanath, D. Tse, and R. Laroia, "Opportunistic beamforming using dumb antennas," *IEEE Trans. Inf. Theory*, vol. 48, no. 6, pp. 1277–1294, Jun. 2002.
- [25] M. Sharif and B. Hassibi, "On the capacity of MIMO broadcast channel with partial side information," *IEEE Trans. Inf. Theory*, vol. 51, no. 2, pp. 506–522, Feb. 2005.
- [26] A. Bayesteh and A. K. Khandani, "On the user selection for MIMO broadcast channels," *IEEE Trans. Inf. Theory*, vol. 54, no. 3, pp. 1086–1107, Mar. 2008.
- [27] T. Yoo and A. Goldsmith, "On the optimality of multi-antenna broadcast scheduling using zero-forcing beamforming," *IEEE JSAC Special Issue on 4 G Wireless Syst.*, vol. 24, no. 3, pp. 528–541, Mar. 2006.
- [28] M. Ebrahimi, M. A. Maddah-Ali, and A. K. Khandani, "Throughput scaling laws for wireless networks with fading channels," *IEEE Trans. Inf. Theory*, vol. 53, no. 11, pp. 4250–4254, Nov. 2007.
- [29] J. Abouei, A. Bayesteh, M. Ebrahimi, and A. K. Khandani, "On the throughput maximization in decentralized wireless networks," *IEEE Trans. Inf. Theory*, Oct. 2008, submitted for publication.
- [30] L. H. Ozarow, S. Shamai, and A. D. Wyner, "Information theoretic considerations for cellular mobile radio," *IEEE Trans. Inf. Theory*, vol. 43, no. 2, pp. 359–378, May 1994.
- [31] P. Gupta and P. R. Kumar, "The capacity of wireless networks," *IEEE Trans. Inf. Theory*, vol. 46, no. 2, pp. 388–404, Mar. 2000.
- [32] M. Franceschetti, O. Dousse, D. Tse, and P. Thiran, "Closing the gap in the capacity of wireless networks via percolation theory," *IEEE Trans. Inf. Theory*, vol. 53, no. 3, pp. 1009–1018, Mar. 2007.
- [33] L.-L. Xie and P. R. Kumar, "A network information theory for wireless communication: Scaling laws and optimal operation," *IEEE Trans. Inf. Theory*, vol. 50, no. 5, pp. 748–767, May 2004.
- [34] M. Gastpar and M. Vetterli, "On the capacity of large Gaussian relay networks," *IEEE Trans. Inf. Theory*, vol. 51, no. 3, pp. 765–779, Mar. 2005.
- [35] O. Lévêque and E. Telatar, "Information theoretic upper bounds on the capacity of large extended ad hoc wireless networks," *IEEE Trans. Inf. Theory*, vol. 51, no. 3, pp. 858–865, Mar. 2005.
- [36] O. Dousse, M. Franceschetti, and P. Thiran, "On the throughput scaling of wireless relay networks," *IEEE Trans. Inf. Theory*, vol. 52, no. 6, pp. 2756–2761, Jun. 2006.
- [37] M. Grossglauser and D. N. C. Tse, "Mobility increases the capacity of ad hoc wireless networks," *IEEE/ACM Trans. Netw.*, vol. 10, pp. 477–486, Aug. 2002.
- [38] S. R. Kulkarni and P. Viswanath, "A deterministic approach to throughput scaling in wireless networks," *IEEE Trans. Inf. Theory*, vol. 50, no. 6, pp. 1041–1049, Jun. 2004.
- [39] R. Gowaikar, B. Hochwald, and B. Hassibi, "Communication over a wireless network with random connections," *IEEE Trans. Inf. Theory*, vol. 52, no. 7, pp. 2857–2871, Jul. 2006.
- [40] R. Gowaikar and B. Hassibi, "On the achievable throughput in two-scale wireless networks," in *Proc. IEEE ISIT*, 2006.
- [41] S. Toupnis and A. J. Goldsmith, "Large wireless networks under fading, mobility, and delay constraints," in *Proc. IEEE Infocom*, Hong Kong, 2004, vol. 1, pp. 609–619.
- [42] F. Xue, L.-L. Xie, and P. R. Kumar, "The transport capacity of wireless networks over fading channels," *IEEE Trans. Inf. Theory*, vol. 51, no. 3, pp. 834–847, Mar. 2005.
- [43] R. Etkin, "Spectrum sharing: Fundamental limits, scaling laws, and self-enforcing protocols," Ph.D., EECS Dep., Univ. Calif., Berkeley, 2006.
- [44] S. Weber, J. G. Andrews, and N. Jindal, "Throughput and transmission capacity of ad hoc networks with channel state information," in *Allerton Conf. Commun., Contr., Comput.*, Monticello, IL, Sep. 2006.
- [45] S. Weber, J. G. Andrews, and N. Jindal, "The effect of fading, channel inversion, and threshold scheduling on ad hoc networks," in *Proc. IEEE Trans. Inf. Theory*, Nov. 2007.
- [46] D. Gesbert and M. Kountouris, "Resource allocation in multicell wireless networks: Some capacity scaling laws," in *Proc. Workshop on Resource Allocation in Wireless Networks (RAWNET 07)*, 2007.
- [47] D. Gesbert and M. Kountouris, "Joint power control and user scheduling in multicell wireless networks: Capacity scaling laws," *IEEE Trans. Inf. Theory*, Sep. 2007.
- [48] N. Jindal, J. G. Andrews, and S. Weber, "Bandwidth-SINR tradeoffs in spatial networks," in *Proc. IEEE Int. Symp. Inf. Theory*, Nice, France, Jun. 2007.
- [49] M. Ebrahimi and A. K. Khandani, "Rate-constrained wireless networks with fading channels: Interference-limited and noise-limited regimes," *IEEE Trans. Inf. Theory*.
- [50] N. Jindal, J. G. Andrews, and S. Weber, "Bandwidth partitioning in decentralized wireless networks," *IEEE Trans. Wireless Commun.*, vol. 7, no. 12, pp. 5408–5419, Dec. 2008.

- [51] N. Jindal, J. G. Andrews, and S. Weber, "Fractional power control for decentralized wireless networks," *IEEE Trans. Wireless Commun.*, vol. 7, no. 12, pp. 5482–5492, Dec. 2008.
- [52] N. Abramson, "The throughput of packet broadcasting channels," *IEEE Trans. Commun.*, vol. 25, no. 1, pp. 117–128, Jan. 1977.
- [53] S. Ghez, S. Verdú, and S. C. Schwartz, "Stability properties of slotted Aloha with multipacket reception capability," *IEEE Trans. Autom. Control*, vol. 33, no. 7, pp. 640–649, Jul. 1988.
- [54] S. Ghez, S. Verdú, and S. C. Schwartz, "Optimal decentralized control in the random access multipacket channel," *IEEE Trans. Autom. Control*, vol. 34, no. 11, pp. 1153–1163, Nov. 1989.
- [55] B. Hajek and T. Loon, "Decentralized dynamic control of a multiaccess broadcast channel," *IEEE Trans. Autom. Control*, vol. 27, no. 3, pp. 559–569, Jun. 1982.
- [56] T. Cui and T. Ho, "On multiple access random medium access control," in *Proc. Int. Symp. Inf. Theory, ISIT09*, Seoul, Korea, Jun. 2009.
- [57] P. Minero, M. Franceschetti, and D. N. C. Tse, "Random access: An information-theoretic perspective," *IEEE Trans. Inf. Theory*, Dec. 2009.
- [58] K. Moshksar, A. Bayesteh, and A. K. Khandani, Coexistence in Wireless Decentralized Networks; Part I: Multiplexing Gain Univ. Waterloo [Online]. Available: http://shannon2.uwaterloo.ca/~kmoshksa/Paper%20I_TR7.pdf
- [59] K. Moshksar, A. Bayesteh, and A. K. Khandani, Coexistence in Wireless Decentralized Networks; Part II: Outage Analysis Univ. Waterloo [Online]. Available: <http://shannon2.uwaterloo.ca/~kmoshksa/>
- [60] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. New York: Wiley, 1991.
- [61] R. G. Gallager, *Information Theory and Reliable Communications*. New York: Wiley, 1968.
- [62] R. M. Dudley, *Real Analysis and Probability*. Cambridge, U.K.: Cambridge Univ. Press, 2002.
- [63] S. M. Kay, *Fundamentals of Statistical Signal Processing, Vol. I: Estimation Theory*. Englewood Cliffs, NJ: Prentice-Hall, 1993.
- [64] S. M. Kay, *Fundamentals of Statistical Signal Processing, Vol. II: Detection Theory*. Englewood Cliffs, NJ: Prentice-Hall, 1998.

Kamyar Moshksar (S10) was born in Shiraz, Iran. He received the B.Sc. and M.Sc. degrees, both in electrical engineering, from Shiraz University and the Sharif University of Technology, respectively, and the Ph.D. degree in February 2011 from the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON, Canada.

He is currently a postdoctoral researcher and lecturer at the University of Waterloo, Waterloo, Ontario, Canada. His interests lie in the broad area of applied probability.

Alireza Bayesteh (S'00) was born in Shiraz, Iran, in 1979. He received the B.Sc. and M.Sc. degrees in electrical engineering from Sharif University of Technology (SUT), Tehran, Iran, in 2000 and 2002, respectively, and the Ph.D. degree in electrical and computer engineering from University of Waterloo, Waterloo, ON, Canada, in 2008.

From 2008 to 2009, he was with the Department of Electrical and Computer Engineering, University of Waterloo, as a Postdoctoral Fellow. In 2009, he joined Research in Motion (RIM), Canada. His research interests are scheduling in wireless networks, especially large-scale networks, MIMO systems, and cooperative communications.

Amir K. Khandani (S'93–M'02) received the M.A.Sc. degree from the University of Tehran, Tehran, Iran, in 1985, and the Ph.D. degree from McGill University, Montreal, QC, Canada, in 1992.

He was a Research Associate with INRS Telecommunications, Quebec University, Montreal, for one year. Since 1993, he has been with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON, Canada, where he is currently a Professor. He currently holds an NSERC Industrial Research Chair (funded by NSERC and Nortel Networks) on Advanced Telecommunications Technologies and a Canada Research Chair (Tier I) on Wireless Systems.

Dr. Khandani is serving as an Associate Editor for the IEEE TRANSACTIONS ON COMMUNICATIONS in the area of coding and communication theory.